From Web to Insight – Automating the Knowledge Pipeline

NLP and Web Analytics Final Project

Arianna Tessari - Student ID: 19322

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Introduction

The objective of this project was to build a multi-step Natural Language Processing (NLP) pipeline capable of collecting, classifying, retrieving, summarizing, and evaluating articles about AI in healthcare. The system integrates several key components, including web scraping, zero-shot classification, Retrieval-Augmented Generation (RAG), summarization, and prompt evaluation. The use of language models throughout the pipeline, in conjunction with tailored prompt engineering strategies, allows for dynamic knowledge extraction and interpretation across a curated dataset.

A diverse range of Python libraries is employed to achieve this goal, each chosen for its strengths in web scraping, natural language processing, data management, or interaction with machine learning models. Here is an overview of the most relevant, along with their main functionalities:

- `requests` and `BeautifulSoup` are used for basic web scraping, enabling HTML parsing and data extraction.
- `selenium` is employed for dynamic web scraping, allowing access to content rendered via JavaScript.
- `pandas` and `numpy` are foundational for data handling and manipulation throughout the pipeline.
- `dotenv` is used to manage environment variables securely, such as API keys.
- `matplotlib`, `Counter`, and `wordcloud` help visualize token distributions and thematic relevance.
- `nltk` is utilized for tokenization, stopword removal, and lemmatization, contributing to text preprocessing.
- `transformers` from Hugging Face provides access to powerful pre-trained models used in zero-shot classification and text generation.
- `sentence_transformers` and `faiss` power the embedding and indexing steps in the information retrieval system.
- `openai` provides access to cutting-edge LLMs such as GPT-4o, used in summarization and QA (Question Answering) tasks.

In [1]:
```
# !pip install requests beautifulsoup4 selenium newspaper3k pandas tqdm
# !pip install "lxml[html_clean]"
```

In [2]:
```
# Importing required libraries

import requests
from bs4 import BeautifulSoup
import pandas as pd
import numpy as np
import os
from dotenv import load_dotenv
from selenium import webdriver
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import urllib.parse
import time
import matplotlib.pyplot as plt
from collections import Counter
from wordcloud import WordCloud
import nltk
from nltk.corpus import stopwords
```

```python
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
from transformers import pipeline
from sentence_transformers import SentenceTransformer, util
import faiss
import textwrap
from openai import OpenAI
import json

load_dotenv()

# Show full columns and rows
pd.set_option('display.max_rows', None)
```

```
c:\Users\arian\OneDrive\Desktop\Natural Language Processing and Web Analytics\myenv_NLP_WebAnalytics\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

1. Web Scraping

The first stage of the pipeline involves gathering a dataset of online articles focused on the intersection of AI and healthcare. To this end, the project targeted the Harvard Gazette, a reliable and reputable news source affiliated with Harvard University. The goal was to collect at least 10 relevant articles related to the theme of **"AI in Healthcare"**, along with their title, URL, publication date, source, and main text content.

To achieve this, two complementary scraping techniques were used: HTTP-based requests and Selenium-driven browser automation. While a standard `requests.get()` call was used to confirm the availability of the base website (`<Response [200]>`), the majority of scraping was handled through `Selenium` , due to the dynamic nature of the Harvard Gazette's search results. This allowed the pipeline to interact with JavaScript-rendered content, simulate user actions like clicking pagination buttons, and open articles in new browser tabs for detailed content extraction.

A custom function, `extract_article_links()` , was implemented to identify and collect URLs pointing to individual article pages. These were filtered to match only those beginning with a known article URL pattern, preventing unrelated links from being included. An auxiliary function, `go_to_next_page()` , enabled the script to navigate through multiple pages of search results to locate enough unique articles.

Each article was then opened in a new browser tab using JavaScript execution via Selenium. Within the article, several different content containers were checked to extract paragraph text, accounting for potential variations in HTML structure. The script also attempted to extract the article's title and date using specific CSS selectors. Articles that failed to provide this metadata were skipped to maintain the integrity of the dataset.

This entire process was run in a loop until 10 successfully parsed articles were collected. The output was stored in a Pandas DataFrame, which was then exported to a CSV file (`ai_healthcare_articles.csv`) for future analysis. A snippet of the collected metadata is shown below.

In [3]:
```python
# page URL
url = 'https://news.harvard.edu/gazette/'

# Making a GET request
r = requests.get(url)

# check status code for response received
# success code - 200
print(r)

# print content of request
# print(r.content)
```

```
<Response [200]>
```

In [4]:
```python
SEARCH_QUERY = "AI in Healthcare"
MAX_ARTICLES_TO_SCRAPE = 10
BASE_URL = 'https://news.harvard.edu/gazette/'

options = Options()
options.add_argument("--start-maximized")
driver = webdriver.Chrome(options=options)

def extract_article_links():
```

```
    links = driver.find_elements(By.CSS_SELECTOR, "div.gsc-webResult.gsc-result div.gs-title a.gs-title")
    urls = []
    for link in links:
        href = link.get_attribute("href")
        if href and href.startswith("https://news.harvard.edu/gazette/story/") and href not in seen_urls:
            urls.append(href)
            seen_urls.add(href)
    return urls

def go_to_next_page():
    try:
        pagination = driver.find_elements(By.CSS_SELECTOR, "div.gsc-cursor-page")
        current_page = driver.find_element(By.CSS_SELECTOR, "div.gsc-cursor-current-page")
        current_index = [i for i, el in enumerate(pagination) if el.text == current_page.text][0]
        if current_index + 1 < len(pagination):
            pagination[current_index + 1].click()
            WebDriverWait(driver, 10).until(
                EC.staleness_of(pagination[current_index + 1])
            )
            time.sleep(2)
            return True
    except Exception as e:
        print("No more pages or error:", e)
    return False

try:
    search_url = f"{BASE_URL}?s={urllib.parse.quote_plus(SEARCH_QUERY)}"
    driver.get(search_url)

    WebDriverWait(driver, 15).until(
        EC.presence_of_all_elements_located((By.CSS_SELECTOR, "div.gsc-webResult.gsc-result"))
    )

    seen_urls = set()
    articles = []
    article_count = 0

    while article_count < MAX_ARTICLES_TO_SCRAPE:
        new_links = extract_article_links()

        for url in new_links:
            if article_count >= MAX_ARTICLES_TO_SCRAPE:
                break

            # Open article in a new tab
            driver.execute_script("window.open('');")
            driver.switch_to.window(driver.window_handles[1])
            driver.get(url)
            time.sleep(2)

            # Scrape article info
            try:
                title = driver.find_element(By.CSS_SELECTOR, ".article-header__title.wp-block-heading").text.stri
            except:
                print(f"Title not found. Skipping: {url}")
                driver.close()
                driver.switch_to.window(driver.window_handles[0])
                continue

            try:
                date = driver.find_element(By.CSS_SELECTOR, "time.entry-date, time.published, time").get_attribut
            except:
                print(f"Date not found. Skipping: {url}")
                driver.close()
                driver.switch_to.window(driver.window_handles[0])
                continue

            content = ""
            containers = [
                "div.entry-content",
                "div.post-content",
                "article.wp-block-post-content",
                "div.article-body",
                "div.the-content"
            ]
```

```
            for container in containers:
                try:
                    paras = driver.find_elements(By.CSS_SELECTOR, f"{container} p")
                    if paras:
                        content = " ".join(p.text for p in paras if p.text.strip())
                        if content:
                            break
                except:
                    continue

            if not content:
                paras = driver.find_elements(By.TAG_NAME, "p")
                content = " ".join(p.text for p in paras if p.text.strip())

            article_count += 1
            print(f"Scraping article {article_count}: {url}")

            articles.append({
                "title": title,
                "url": url,
                "date": date,
                "source": "Harvard Gazette",
                "content": content
            })

            # Close article tab and switch back to search results
            driver.close()
            driver.switch_to.window(driver.window_handles[0])

        if article_count >= MAX_ARTICLES_TO_SCRAPE:
            break

        if not go_to_next_page():
            print("No more pages.")
            break

    df = pd.DataFrame(articles)

finally:
    driver.quit()
    print("Driver closed.")
```

```
Scraping article 1: https://news.harvard.edu/gazette/story/2025/03/how-ai-is-transforming-medicine-healthcare/
Scraping article 2: https://news.harvard.edu/gazette/story/2019/02/in-health-care-ai-offers-promise-and-hype/
Scraping article 3: https://news.harvard.edu/gazette/story/2020/10/ethical-concerns-mount-as-ai-takes-bigger-decis
ion-making-role/
Scraping article 4: https://news.harvard.edu/gazette/story/2024/09/new-ai-tool-can-diagnose-cancer-guide-treatment
-predict-patient-survival/
Scraping article 5: https://news.harvard.edu/gazette/story/2020/11/risks-and-benefits-of-an-ai-revolution-in-medic
ine/
Scraping article 6: https://news.harvard.edu/gazette/story/2024/09/using-ai-to-repurpose-existing-drugs-for-treatm
ent-of-rare-diseases/
Scraping article 7: https://news.harvard.edu/gazette/story/2025/02/should-ai-be-used-in-end-of-life-medical-decisi
ons-a-i-artificial-intelligence/
Scraping article 8: https://news.harvard.edu/gazette/story/2023/11/harvard-symposium-looks-at-ai-and-oral-health/
Scraping article 9: https://news.harvard.edu/gazette/story/2024/11/cutting-through-the-fog-of-long-covid/
Scraping article 10: https://news.harvard.edu/gazette/story/newsplus/new-journal-podcast-take-a-closer-look-at-art
ificial-intelligence-in-medicine/
Driver closed.
```

In [5]:

```
df
```

Out[5]:

	title	url	date	source	content
0	Machine healing	https://news.harvard.edu/gazette/story/2025/03...	2025-03-20	Harvard Gazette	Photo illustrations by Judy Blomquist/Harvard ...
1	The algorithm will see you now	https://news.harvard.edu/gazette/story/2019/02...	2019-02-28	Harvard Gazette	Ashley Nunes, a fellow at Harvard Law School, ...
2	Great promise but potential for peril	https://news.harvard.edu/gazette/story/2020/10...	2020-10-26	Harvard Gazette	Illustration by Ben Boothman Christina Pazzane...
3	New AI tool can diagnose cancer, guide treatme...	https://news.harvard.edu/gazette/story/2024/09...	2024-09-04	Harvard Gazette	Video: Wirestock/Creatas Video+/Getty Images P...
4	AI revolution in medicine	https://news.harvard.edu/gazette/story/2020/11...	2020-11-11	Harvard Gazette	Illustration by Ben Boothman Alvin Powell Harv...
5	Using AI to repurpose existing drugs for treat...	https://news.harvard.edu/gazette/story/2024/09...	2024-09-25	Harvard Gazette	Identifies possible therapies for thousands of...
6	It's inoperable cancer. Should AI make call ab...	https://news.harvard.edu/gazette/story/2025/02...	2025-02-10	Harvard Gazette	Rebecca Weintraub Brendel, director of Harvard...
7	AI may be just what the dentist ordered	https://news.harvard.edu/gazette/story/2023/11...	2023-11-29	Harvard Gazette	Kat J. McAlpine Harvard Correspondent Dental p...
8	Cutting through the fog of long COVID	https://news.harvard.edu/gazette/story/2024/11...	2024-11-08	Harvard Gazette	MGB Communications While earlier diagnostic st...
9	New journal, podcast take a closer look at art...	https://news.harvard.edu/gazette/story/newsplu...	2023-04-10	Harvard Gazette	iStock/ipopba Artificial intelligence (AI) has...

In [6]:
```
df.to_csv("ai_healthcare_articles.csv", index=False)
```

To verify the relevance and quality of the data, each article was printed with all its metadata. This allowed a manual check of whether the content aligned with the search query and whether the scraping logic was successfully extracting useful information.

In [7]:
```
df = pd.read_csv("ai_healthcare_articles.csv")

# Print each article with title, date, source, URL, and content
for idx, row in df.iterrows():
    print(f"\n--- ARTICLE {idx+1} ---")
    print(f"Title   : {row['title']}")
    print(f"Date    : {row['date']}")
    print(f"Source  : {row['source']}")
    print(f"URL     : {row['url']}\n")
    print(f"Content :\n{row['content']}")
    print("\n" + "-"*80)
```

--- ARTICLE 1 ---
Title : Machine healing
Date : 2025-03-20
Source : Harvard Gazette
URL : https://news.harvard.edu/gazette/story/2025/03/how-ai-is-transforming-medicine-healthcare/

Content :
Photo illustrations by Judy Blomquist/Harvard Staff Alvin Powell Harvard Staff Writer When Adam Rodman was a second-year medical student in the 2000s, he visited the library for a patient whose illness had left doctors stumped. Rodman searched the catalog, copied research papers, and shared them with the team. “It made a big difference in that patient’s care,” Rodman said. “Everyone said, ‘This is so great. This is evidence-based medicine.’ But it took two hours. I can do that today in 15 seconds.” Rodman, now an assistant professor at Harvard Medical School and a doctor at Beth Israel Deaconess Medical Center, these days carries a medical library in his pocket — a smartphone app created after the release of the large language model ChatGPT in 2022. OpenEvidence — developed in part by Medical School faculty — allows him to query specific diseases and symptoms. It searches the medical literature, drafts a summary of findings, and lists the most important sources for further reading, providing answers while Rodman is still face-to-face with his patient. “We say, ‘Wow, the technology is really powerful.’ But what do we do with it to actually change things?” Artificial intelligence in various forms has been used in medicine for decades — but not like this. Experts predict that the adoption of large language models will reshape medicine. Some compare the potential impact with the decoding of the human genome, even the rise of the internet. The impact is expected to show up in doctor-patient interactions, physicians’ paperwork load, hospital and physician practice administration, medical research, and medical education. Most of these effects are likely to be positive, increasing efficiency, reducing mistakes, easing the nationwide crunch in primary care, bringing data to bear more fully on decision-making, reducing administrative burdens, and creating space for longer, deeper person-to-person interactions. Adam Rodman, assistant professor at Harvard Medical School and physician at Beth Israel Deaconess Medical Center But there are serious concerns, too. Current data sets too often reflect societal biases that reinforce gaps in access and quality of care for disadvantaged groups. Without correction, these data have the potential to cement existing biases into ever-more-powerful AI that will increasingly influence how healthcare operates. Another important issue, experts say, is that AIs remain prone to “hallucination,” making up “facts” and presenting them as if they are real. Then there’s the danger that medicine won’t be bold enough. The latest AI has the potential to remake health care top to bottom, but only if given a chance. The wrong priorities — too much deference to entrenched interests, a focus on money instead of health — could easily reduce the AI “revolution” to an underwhelming exercise in tinkering around the edges. “I think we’re in this weird space,” Rodman said. “We say, ‘Wow, the technology is really powerful.’ But what do we do with it to actually change things? My worry, as both a clinician and a researcher, is that if we don’t think big, if we don’t try to rethink how we’ve organized medicine, things might not change that much.” Five years ago, when asked about AI in healthcare, Isaac Kohane responded with frustration. Teenagers tapping away on social media apps were better equipped than many doctors. The situation today couldn’t be more different, he says. Kohane, chair of the Medical School’s Department of Biomedical Informatics and editor-in-chief of the New England Journal of Medicine’s new AI initiative, describes the abilities of the latest models as “mind boggling.” To illustrate the point, he recalled getting an early look at OpenAI’s GPT-4. He tested it with a complex case — a child born with ambiguous genitalia — that might have stymied even an experienced endocrinologist. Kohane asked GPT-4 about genetic causes, biochemical pathways, next steps in the workup, even what to tell the child’s parents. It aced the test. “This large language model was not trained to be a doctor; it’s just trained to predict the next word,” Kohane said. “It could speak as coherently about wine pairings with a vegetarian menu as diagnose a complex patient. It was truly a quantum leap from anything that anybody in computer science who was honest with themselves would have predicted in the next 10 years.” Isaac Kohane, chairman of Harvard Medical School’s Department of Biomedical Informatics and editor-in-chief of the New England Journal of Medicine’s new AI journal And none too soon. The U.S. healthcare system, long criticized as costly, inefficient, and inordinately focused on treatment over prevention, has been showing cracks. Kohane, recalling a faculty member new to the department who couldn’t find a primary care physician, is tired of seeing them up close. “The medical system, which I have long said is broken, is broken in extremely obvious ways in Boston,” he said. “People worry about equity problems with AI. I’m here to say we have a huge equity problem today. Unless you’re well connected and are willing to pay literally thousands of extra dollars for concierge care, you’re going to have trouble finding a timely primary care visit.” Early worries that AI would replace physicians have yielded to the realization that the system needs both AI and its human workforce, Kohane said. Teaming nurse practitioners and physician assistants with AI is one among several promising scenarios. “It is no longer a conversation about, ‘Will AI replace doctors,’ so much as, ‘Will AI, with a set of clinicians who may not look like the clinicians that we’re used to, firm up the tottering edifice that is organized medicine?’” How LLMs were rolled out — to everyone at once — accelerated their adoption, Kohane says. Doctors immediately experimented with eye-glazing yet essential tasks, like writing prior authorization requests to insurers explaining the necessity of specific, usually expensive, treatments. “People just did it,” Kohane said. “Doctors were tweeting back and forth about all the time they were saving.” Patients did it too, seeking virtual second opinions, like the child whose recurring pain was misdiagnosed by 17 doctors over three years. In the widely publicized case, the boy’s mother entered his medical notes into ChatGPT, which suggested a condition no doctor had mentioned: tethered cord syndrome, in which the spinal cord binds inside of the backbone. When the patient moves, rather than sliding smoothly, the spinal cord stretches, causing pain. The diagnosis was confirmed by a neurosurgeon, who then corrected the anatomic anomaly. One of the perceived benefits of employing AI in the clinic, of course, is to make doctors better the first time around. Greater, faster access to case histories, suggested diagnoses, and other data is expected to improve physician performance. But plenty of work remains, a recent study shows. Research published in JAMA Network Open in October compared diagnoses delivered by an individual doctor, a doctor using an LLM diagnostic tool, and an LLM alone. The results were surprising, showing an insignificant improvement in accuracy for the physicians using the LLM — 76 percent versus 74 percent for the solitary physician. More surprisingly, the LLM by itself did best, scoring 16 percentage points higher than physicians alone. Rodman, one of the paper’s senior authors, said it’s tempting to conclude that LLMs aren’t that helpful for doctors, but he insisted that it’s important to look deeper at the findings. Only 10 percent of the physicians, he said, were experienced LLM users before the study — which took place in 2023— and the rest received only basic training. Consequently, when Rodman lat

er looked at the transcripts, most used the LLMs for basic fact retrieval. “The best way a doctor could use it now is for a second opinion, to second-guess themselves when they have a tricky case,” he said. “How could I be wrong? What am I missing? What other questions should I ask? Those are the ways, we know from psychological literature, that complement how humans think.” Among the other potential benefits of AI is the chance to make medicine safer, according to David Bates, co-director of the Center for Artificial Intelligence and Bioinformatics Learning Systems at Mass General Brigham. A recent study by Bates and colleagues showed that as many as one in four visits to Massachusetts hospitals results in some kind of patient harm. Many of those incidents trace back to adverse drug events. “AI should be able to look for medication-related issues and identify them much more accurately than we’re able to do right now,” said Bates, who is also a professor of medicine at the Medical School and of health policy and management at the Harvard T.H. Chan School of Public Health. David Bates, co-director of the Center for Artificial Intelligence and Bioinformatics Learning Systems at Mass General Brigham Another opportunity stems from AI’s growing competence in a mundane area: notetaking and summarization, according to Bernard Chang, dean for medical education at the Medical School. Systems for “ambient documentation” will soon be able to listen in on patient visits, record everything that is said and done, and generate an organized clinical note in real time. When symptoms are discussed, the AI can suggest diagnoses and courses of treatment. Later, the physician can review the summary for accuracy. Automation of notes and summaries would benefit healthcare workers in more than one way, Chang said. It would ease doctors’ paperwork load, often cited as a cause of burnout, and it would reset the doctor-patient relationship. One of patients’ biggest complaints about office visits is the physician sitting at the computer, asking questions and recording the answers. Freed from the note-taking process, doctors could sit face-to-face with patients, opening a path to stronger connections.   “It’s not the most magical use of AI,” Chang said. “We’ve all seen AI do something and said, ‘Wow, that’s amazing.’ This is not one of those things. But this program is being piloted at different ambulatory practices across the country and the early results are very promising. Physicians who feel overburdened and burnt out are starting to say, ‘You know what, this tool is going to help me.’” Bernard Chang, Harvard Medical School Dean for Medical Education For all their power, LLMs are not ready to be left alone. “The technology is not good enough to have that safety level where you don’t need a knowledgeable human,” Rodman said. “I can understand where it might have gone aground. I can take a step further with the diagnosis. I can do that because I learned the hard way. In residency you make a ton of mistakes, but you learn from those mistakes. Our current system is incredibly suboptimal but it does train your brain. When people in medical school interact with things that can automate those processes — even if they’re, on average, better than humans — how are they going to learn?” Doctors and scientists also worry about bad information. Pervasive data bias stems from biomedicine’s roots in wealthy Western nations whose science was shaped by white men studying white men, says Leo Celi, an associate professor of medicine and a physician in the Division of Pulmonary, Critical Care and Sleep Medicine at Beth Israel Deaconess Medical Center. Leo Celi, associate professor of medicine and a physician in Beth Israel Deaconess Medical Center’s Division of Pulmonary, Critical Care and Sleep Medicine “You need to understand the data before you can build artificial intelligence,” Celi said. “That gives us a new perspective of the design flaws of legacy systems for healthcare delivery, legacy systems for medical education. It becomes clear that the status quo is so bad — we knew it was bad and we’ve come to accept that it is a broken system — that all the promises of AI are going bust unless we recode the world itself.” Celi cited research on disparities in care between English-speaking and non-English speaking patients hospitalized with diabetes. Non-English speakers are woken up less frequently for blood sugar checks, raising the likelihood that changes will be missed. That impact is hidden, however, because the data isn’t obviously biased, only incomplete, even though it still contributes to a disparity in care. “They have one or two blood-sugar checks compared to 10 if you speak English well,” he said. “If you average it, the computers don’t see that this is a data imbalance. There’s so much missing context that experts may not be aware of what we call ‘data artifacts.’ This arises from a social patterning of the data generation process.” Bates offered additional examples, including a skin cancer device that does a poor job detecting cancer on highly pigmented skin and a scheduling algorithm that wrongly predicted Black patients would have higher no-show rates, leading to overbooking and longer wait times. “Most clinicians are not aware that every medical device that we have is, to a certain degree, biased,” Celi said. “They don’t work well across all groups because we prototype them and we optimize them on, typically, college-age, white, male students. They were not optimized for an ICU patient who is 80 years old and has all these comorbidities, so why is there an expectation that the numbers they represent are objective ground truths?” The exposure of deep biases in legacy systems presents an opportunity to get things right, Celi said. Accordingly, more researchers are pushing to ensure that clinical trials enroll diverse populations from geographically diverse locations. One example is Beth Israel’s MIMIC database, which reflects the hospital’s diverse patient population. The tool, overseen by Celi, offers investigators de-identified electronic medical records — notes, images, test results — in an open-source format. It has been used in 10,000 studies by researchers all around the world and is set to expand to 14 additional hospitals, he said. As in the clinic, AI models used in the lab aren’t perfect, but they are opening pathways that hold promise to greatly accelerate scientific progress. “They provide instant insights at the atomic scale for some molecules that are still not accessible experimentally or that would take a tremendous amount of time and effort to generate,” said Marinka Zitnik, an associate professor of biomedical informatics at the Medical School. “These models provide in-silico predictions that are accurate, that scientists can then build upon and leverage in their scientific work. That, to me, just hints at this incredible moment that we are in.” ”What is becoming increasingly important is to develop reliable, faithful benchmarks or techniques that allow us to evaluate how well the outputs of AI models behave in the real world.” Zitnik’s lab recently introduced Procyon, an AI model aimed at closing knowledge gaps around protein structures and their biological roles. Until recently, it has been difficult for scientists to understand a protein’s shape — how the long molecules fold and twist onto themselves in three dimensions. This is important because the twists and turns expose portions of the molecule and hide others, making those sites easier or harder for other molecules to interact with, which affects the molecule’s chemical properties. Marinka Zitnik, assistant professor of biomedical informatics Today, predicting a protein’s shape — down to nearly every atom — from its known sequence of amino acids is feasible, Zitnik said. The major challenge is linking those structures to their functions and phenotypes across various biological settings and diseases. About 20 percent of human proteins have poorly defined functions, and an overwhelming share of research — 95 percent — is devoted to just 5,000 well-studied proteins. “We are addressing this gap by connecting molecular sequences and structures with functional annotations to predict protein phenotypes, helping move the field closer to being able to in-silico predict functions for each protein,” Zitnik said. A long-term goal for AI in the lab is the development of “AI scientists” that function as research assistants, with access to the entire body of scientifi

c literature, the ability to integrate that knowledge with experimental results, and the capacity to suggest next
steps. These systems could evolve into true collaborators, Zitnik said, noting that some models have already gener
ated simple hypotheses. Her lab used Procyon, for example, to identify domains in the maltase glucoamylase protein
that bind miglitol, a drug used to treat Type 2 diabetes. In another project, the team showed that Procyon could f
unctionally annotate poorly characterized proteins implicated in Parkinson’s disease. The tool’s broad range of ca
pabilities is possible because it was trained on massive experimental data sets and the entire scientific literatu
re, resources far exceeding what humans can read and analyze, Zitnik said. The classroom comes before the lab, and
the AI dynamic of flexibility, innovation, and constant learning is also being applied to education. The Medical S
chool has introduced a course dealing with AI in healthcare; added a Ph.D. track on AI in medicine; is planning a
“tutor bot” to provide supplemental material beyond lectures; and is developing a virtual patient on which student
s can practice before their first nerve-wracking encounter with the real thing. Meanwhile, Rodman is leading a ste
ering group on the use of generative AI in medical education. These initiatives are a good start, he said. Still,
the rapid evolution of AI technology makes it difficult to prepare students for careers that will span 30 years.
“The Harvard view, which is my view as well, is that we can give people the basics, but we just have to encourage
agility and prepare people for a future that changes rapidly,” Rodman said. “Probably the best thing we can do is
prepare people to expect the unexpected.”

--- ARTICLE 2 ---
Title   : The algorithm will see you now
Date   : 2019-02-28
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/2019/02/in-health-care-ai-offers-promise-and-hype/

Content :
Ashley Nunes, a fellow at Harvard Law School, speaks during The Harvard Global Health Institute conference on AI i
n health care. Photos by Kris Snibbe/Harvard Staff Photographer Alvin Powell Harvard Staff Writer In 2016, a Googl
e team announced it had used artificial intelligence to diagnose diabetic retinopathy — one of the fastest-growing
causes of blindness — as well as trained eye doctors could. In December 2018, Microsoft and the pharmaceutical gia
nt Novartis announced a partnership to develop an AI-powered digital health tool to be deployed against one of hum
anity’s oldest scourges — leprosy, which still afflicts 200,000 new patients annually. Around the world, artificia
l intelligence is being touted as the next big thing in health care, and a potential game-changer for billions liv
ing in regions where medical infrastructure is inadequate and doctors and nurses scarce. Despite that potential, s
ome fear that too-rosy views of AI’s promise will lead to disappointment and, worse, rob scarce health care dollar
s from desperately needed investments in existing medical infrastructure. Experts in AI and global health gathered
at Harvard this week to survey the complex artificial intelligence-global health landscape, examine what’s being d
one today and what’s promised for tomorrow, and try to separate reality from fantasy when it comes to AI’s potenti
al impact on global health. “The promise of AI is enormous, but the challenges are often glossed over,” said Ashis
h Jha, faculty director of Harvard Global Health Institute (HGHI). “We have this sense that they’ll somehow take c
are of themselves. And we know they will not.” About 50 experts in the use of artificial intelligence and other di
gital health care technologies gathered at Loeb House for an all-day symposium on “Hype vs. Reality: The Role of A
I in Global Health,” sponsored by HGHI. It featured talks by representatives from academia, health care, and indus
try, including Google AI and the pharmaceutical giant Novartis. Principals also spoke for startups Aindra Systems,
which uses AI to rapidly analyze pap smears for evidence of cervical cancer, and Ubenwa, which has developed an AI
-based system to analyze infant cries to detect birth asphyxia, a leading killer of children under 5 that’s tradit
ionally diagnosed through time-consuming blood tests and lab analyses. Despite the symposium’s cautionary tone, se
veral speakers described AI’s potentially far-reaching effects on global health. Image analysis, a key aspect of d
isease screening and diagnosis, is an area ripe for rapid change. Lily Peng, product manager for Google AI, descri
bed Google’s work on diabetic retinopathy. The company’s researchers used a “deep learning” algorithm that reviewe
d thousands of images of both normal and damaged eyes and developed a screening tool for the condition — a side ef
fect of poorly controlled diabetes — that performed as well as trained ophthalmologists. That tool, and similar on
es for other conditions, can provide valuable screening and referral services in rural areas that have little medi
cal infrastructure. Their promise is of a future where initial screening can be guided by community health workers
— far more numerous than trained physicians — who can then refer those who test positive to clinics and hospitals
staffed with better-trained medical personnel. But the question remained: How effectively would such interventions
perform in the field? During a Q&A session, one participant described a device developed by an engineer in Cameroo
n to diagnose cardiac problems in remote areas. The device was designed to be used by rural nurses who could send
results to city-based cardiologists for review. Problems occurred, however, because the nurses entered data in the
wrong fields, spotty internet connections made it hard to transmit the data, and finally, when the information mad
e it to the city, the cardiologists were often too busy to examine it. Adam Landman, chief information officer at
Brigham and Women’s Hospital and an associate professor of emergency medicine at Harvard Medical School, said it’s
important that AI not be viewed as a solution in search of a problem, but that the health care outcomes be conside
red first and AI considered as one tool among others to address it. “The key with any technology — and AI is just
one of them — is it needs to actually solve a health care [problem],” Landman said. “The need [should] drive the u
se of it and not the technology drive the use. They may not need AI to start. In fact, they may not need any techn
ology to start. They may need more people, more health care workers. There may be other things to invest in before
technology, or technology may be part of that solution.” About 50 experts in the use of artificial intelligence an
d other digital health care technologies gathered at Loeb House for the all-day symposium. Ann Aerts, head of Nova
rtis Foundation, said severe shortages of doctors and nurses in some parts of the world demand innovative approach
es to health care. Technology has already transformed the way medicine is practiced in some resource-poor setting
s. In Ghana, a telehealth pilot has evolved into a national telehealth program that keeps a center staffed 24 hour
s a day, seven days a week, with doctors, nurses, and health workers. The program isn’t a substitute for skilled m
edical care, but Aerts said that about 70 percent of all health care problems can be solved over the phone. The pr
ogram highlights how innovative solutions can be found to the health care shortage that is a fact of life in many

parts of the world. “This calls for innovating the way we deliver health services, for transforming our health systems,” Aerts said. “We cannot go back to the old model. It doesn’t work. … Let’s look at how we can reimagine the way we deliver health services.” John Halamka, chief information officer at Beth Israel Deaconess Medical Center and the International Healthcare Innovation Professor of Emergency Medicine at Harvard Medical School, said that AI has the potential not only to meet needs where care is scarce, but also to improve care and reduce errors where doctors and nurses are present. He told of a woman who visited a doctor in northern India because of abdominal pain. She was incorrectly diagnosed and paid for repeated testing and physician visits. Ultimately, she was forced her to sell her house and cow, impoverishing her family. She was finally diagnosed correctly with abdominal tuberculosis, a condition that might rarely be seen in the U.S., but ought to have been considered much sooner in India, which has 1.3 million active cases of tuberculosis, Halamka said. Halamka consulted on another case, in Bakersfield, California, that shows the problem is not restricted to the developing world. A young woman was arrested for running naked through a Walmart parking lot at 3 a.m. When Halamka called to talk to the emergency room physician, he was told that the young woman’s urine had tested positive for cannabis, so he had discharged her. “And of course all of us in this room know that cannabis consumption results in young women running naked through parking lots at 3 in the morning — never,” Halamka said. “We flew her to Logan [International Airport], brought her to Beth Israel Deaconess … [and she had] the worst case of meningitis any of us had ever seen. Again thinking back on AI, if we had [it trained by the experiences of] a million young women with altered mental status, how many would include cannabis as the primary cause? Probably, again, none. But in a person who lives with a lot of other 20-year-olds who have lots of infectious diseases, what are the chances of meningitis? Actually, pretty significant.” Halamka said that artificial intelligence and machine learning has progressed rapidly since the late 1990s, when it took six months of work to get AI to recognize a giraffe as a mammal. Today, he said, designing an AI algorithm itself is relatively easy, but algorithms have to be trained on large amounts of data from past cases, and it can be challenging to ensure that data is available, appropriate, and unbiased. “Gathering and curating good-quality data, unbiased data, and using it to train these algorithms is still a challenge,” Halamka said. “The quality of our data is poor. We get social security numbers wrong 11 percent of the time. We even get gender wrong 3 percent of the time. Men have babies, women have prostatectomies. We need to work on curated data sets if we’re going to be successful with AI.” Designers of AI health care applications, Aerts and other speakers said, should also guard against AI becoming a benefit solely for the rich, a factor that increases inequality instead of helping close the gap between haves and have-nots. Ashley Nunes, a fellow at Harvard Law School, said there’s a perception that AI will cut health care costs, making it more affordable for the poor, but there’s no guarantee that will happen. Nunes pointed to self-driving cars, which are projected to make roads safer. Accident data show the poor suffer the most physical and economic harm from road accidents, so hypothetically they could see the most benefit. But the added cost of the driverless technology means it will likely disproportionately benefit wealthier drivers, highlighting how important it is to ensure that tech solutions in health care benefit everyone. Jha said he looks forward to a time when AI is just another tool in the health care toolbox. “The whole idea of digital health reminds me that about 20 years ago we used to talk about e-commerce,” he said. “We don’t talk about e-commerce anymore; we just talk about commerce.”

--- ARTICLE 3 ---
Title   : Great promise but potential for peril
Date   : 2020-10-26
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/2020/10/ethical-concerns-mount-as-ai-takes-bigger-decision-making-role/

Content :
Illustration by Ben Boothman Christina Pazzanese Harvard Staff Writer Second in a four-part series that taps the expertise of the Harvard community to examine the promise and potential pitfalls of the rising age of artificial intelligence and machine learning, and how to humanize them. For decades, artificial intelligence, or AI, was the engine of high-level STEM research. Most consumers became aware of the technology’s power and potential through internet platforms like Google and Facebook, and retailer Amazon. Today, AI is essential across a vast array of industries, including health care, banking, retail, and manufacturing. But its game-changing promise to do things like improve efficiency, bring down costs, and accelerate research and development has been tempered of late with worries that these complex, opaque systems may do more societal harm than economic good. With virtually no U.S. government oversight, private companies use AI software to make determinations about health and medicine, employment, creditworthiness, and even criminal justice without having to answer for how they’re ensuring that programs aren’t encoded, consciously or unconsciously, with structural biases. Its growing appeal and utility are undeniable. Worldwide business spending on AI is expected to hit $50 billion this year and $110 billion annually by 2024, even after the global economic slump caused by the COVID-19 pandemic, according to a forecast released in August by technology research firm IDC. Retail and banking industries spent the most this year, at more than $5 billion each. The company expects the media industry and federal and central governments will invest most heavily between 2018 and 2023 and predicts that AI will be “the disrupting influence changing entire industries over the next decade.” “Virtually every big company now has multiple AI systems and counts the deployment of AI as integral to their strategy,” said Joseph Fuller, professor of management practice at Harvard Business School, who co-leads Managing the Future of Work, a research project that studies, in part, the development and implementation of AI, including machine learning, robotics, sensors, and industrial automation, in business and the work world. Early on, it was popularly assumed that the future of AI would involve the automation of simple repetitive tasks requiring low-level decision-making. But AI has rapidly grown in sophistication, owing to more powerful computers and the compilation of huge data sets. One branch, machine learning, notable for its ability to sort and analyze massive amounts of data and to learn over time, has transformed countless fields, including education. Firms now use AI to manage sourcing of materials and products from suppliers and to integrate vast troves of information to aid in strategic decision-making, and because of its capacity to process data so quickly, AI tools are helping to minimize time in the pricey trial-and-error of product development — a critical advance for an industry like pharmaceuticals, where it costs $1 billion to bring a new pill to market, Fuller said. Health care experts see many possible uses for AI, including with b

illing and processing necessary paperwork. And medical professionals expect that the biggest, most immediate impact will be in analysis of data, imaging, and diagnosis. Imagine, they say, having the ability to bring all of the medical knowledge available on a disease to any given treatment decision. In employment, AI software culls and processes resumes and analyzes job interviewees' voice and facial expressions in hiring and driving the growth of what's known as "hybrid" jobs. Rather than replacing employees, AI takes on important technical tasks of their work, like routing for package delivery trucks, which potentially frees workers to focus on other responsibilities, making them more productive and therefore more valuable to employers. "It's allowing them to do more stuff better, or to make fewer errors, or to capture their expertise and disseminate it more effectively in the organization," said Fuller, who has studied the effects and attitudes of workers who have lost or are likeliest to lose their jobs to AI. — Michael Sandel, political philosopher and Anne T. and Robert M. Bass Professor of Government Though automation is here to stay, the elimination of entire job categories, like highway toll-takers who were replaced by sensors because of AI's proliferation, is not likely, according to Fuller. "What we're going to see is jobs that require human interaction, empathy, that require applying judgment to what the machine is creating [will] have robustness," he said. While big business already has a huge head start, small businesses could also potentially be transformed by AI, says Karen Mills '75, M.B.A. '77, who ran the U.S. Small Business Administration from 2009 to 2013. With half the country employed by small businesses before the COVID-19 pandemic, that could have major implications for the national economy over the long haul. Rather than hamper small businesses, the technology could give their owners detailed new insights into sales trends, cash flow, ordering, and other important financial information in real time so they can better understand how the business is doing and where problem areas might loom without having to hire anyone, become a financial expert, or spend hours laboring over the books every week, Mills said. One area where AI could "completely change the game" is lending, where access to capital is difficult in part because banks often struggle to get an accurate picture of a small business's viability and creditworthiness. "It's much harder to look inside a business operation and know what's going on" than it is to assess an individual, she said. Information opacity makes the lending process laborious and expensive for both would-be borrowers and lenders, and applications are designed to analyze larger companies or those who've already borrowed, a built-in disadvantage for certain types of businesses and for historically underserved borrowers, like women and minority business owners, said Mills, a senior fellow at HBS. But with AI-powered software pulling information from a business's bank account, taxes, and online bookkeeping records and comparing it with data from thousands of similar businesses, even small community banks will be able to make informed assessments in minutes, without the agony of paperwork and delays, and, like blind auditions for musicians, without fear that any inequity crept into the decision-making. "All of that goes away," she said. Not everyone sees blue skies on the horizon, however. Many worry whether the coming age of AI will bring new, faster, and frictionless ways to discriminate and divide at scale. "Part of the appeal of algorithmic decision-making is that it seems to offer an objective way of overcoming human subjectivity, bias, and prejudice," said political philosopher Michael Sandel, Anne T. and Robert M. Bass Professor of Government. "But we are discovering that many of the algorithms that decide who should get parole, for example, or who should be presented with employment opportunities or housing … replicate and embed the biases that already exist in our society." — Karen Mills, senior fellow at the Business School and head of the U.S. Small Business Administration from 2009 to 2013 AI presents three major areas of ethical concern for society: privacy and surveillance, bias and discrimination, and perhaps the deepest, most difficult philosophical question of the era, the role of human judgment, said Sandel, who teaches a course in the moral, social, and political implications of new technologies. "Debates about privacy safeguards and about how to overcome bias in algorithmic decision-making in sentencing, parole, and employment practices are by now familiar," said Sandel, referring to conscious and unconscious prejudices of program developers and those built into datasets used to train the software. "But we've not yet wrapped our minds around the hardest question: Can smart machines outthink us, or are certain elements of human judgment indispensable in deciding some of the most important things in life?" Panic over AI suddenly injecting bias into everyday life en masse is overstated, says Fuller. First, the business world and the workplace, rife with human decision-making, have always been riddled with "all sorts" of biases that prevent people from making deals or landing contracts and jobs. When calibrated carefully and deployed thoughtfully, resume-screening software allows a wider pool of applicants to be considered than could be done otherwise, and should minimize the potential for favoritism that comes with human gatekeepers, Fuller said. Sandel disagrees. "AI not only replicates human biases, it confers on these biases a kind of scientific credibility. It makes it seem that these predictions and judgments have an objective status," he said. In the world of lending, algorithm-driven decisions do have a potential "dark side," Mills said. As machines learn from data sets they're fed, chances are "pretty high" they may replicate many of the banking industry's past failings that resulted in systematic disparate treatment of African Americans and other marginalized consumers. "If we're not thoughtful and careful, we're going to end up with redlining again," she said. A highly regulated industry, banks are legally on the hook if the algorithms they use to evaluate loan applications end up inappropriately discriminating against classes of consumers, so those "at the top levels" in the field are "very focused" right now on this issue, said Mills, who closely studies the rapid changes in financial technology, or "fintech." "They really don't want to discriminate. They want to get access to capital to the most creditworthy borrowers," she said. "That's good business for them, too." Given its power and expected ubiquity, some argue that the use of AI should be tightly regulated. But there's little consensus on how that should be done and who should make the rules. Thus far, companies that develop or use AI systems largely self-police, relying on existing laws and market forces, like negative reactions from consumers and shareholders or the demands of highly-prized AI technical talent to keep them in line. "There's no businessperson on the planet at an enterprise of any size that isn't concerned about this and trying to reflect on what's going to be politically, legally, regulatorily, [or] ethically acceptable," said Fuller. Firms already consider their own potential liability from misuse before a product launch, but it's not realistic to expect companies to anticipate and prevent every possible unintended consequence of their product, he said. Few think the federal government is up to the job, or will ever be. "The regulatory bodies are not equipped with the expertise in artificial intelligence to engage in [oversight] without some real focus and investment," said Fuller, noting the rapid rate of technological change means even the most informed legislators can't keep pace. Requiring every new product using AI to be prescreened for potential social harms is not only impractical, but would create a huge drag on innovation. — Jason Furman, a professor of the practice of economic policy at the Kennedy School and a former top economic adviser to President Barack Obama Jason Furman, a professor of the practice of economic policy at Harvard Kennedy School, agrees that government regulators need "a much better technical understanding of artificial intelligence to do that job well," but says they could do it. Existing bodies like the National H

ighway Transportation Safety Association, which oversees vehicle safety, for example, could handle potential AI is
sues in autonomous vehicles rather than a single watchdog agency, he said. “I wouldn’t have a central AI group tha
t has a division that does cars, I would have the car people have a division of people who are really good at AI,”
said Furman, a former top economic adviser to President Barack Obama. Though keeping AI regulation within industri
es does leave open the possibility of co-opted enforcement, Furman said industry-specific panels would be far more
knowledgeable about the overarching technology of which AI is simply one piece, making for more thorough oversigh
t. While the European Union already has rigorous data-privacy laws and the European Commission is considering a fo
rmal regulatory framework for ethical use of AI, the U.S. government has historically been late when it comes to t
ech regulation. “I think we should’ve started three decades ago, but better late than never,” said Furman, who thi
nks there needs to be a “greater sense of urgency” to make lawmakers act. Business leaders “can’t have it both way
s,” refusing responsibility for AI’s harmful consequences while also fighting government oversight, Sandel maintai
ns. “The problem is these big tech companies are neither self-regulating, nor subject to adequate government regul
ation. I think there needs to be more of both,” he said, later adding: “We can’t assume that market forces by them
selves will sort it out. That’s a mistake, as we’ve seen with Facebook and other tech giants.” Last fall, Sandel t
aught “Tech Ethics,” a popular new Gen Ed course with Doug Melton, co-director of Harvard’s Stem Cell Institute. A
s in his legendary “Justice” course, students consider and debate the big questions about new technologies, everyt
hing from gene editing and robots to privacy and surveillance. “Companies have to think seriously about the ethica
l dimensions of what they’re doing and we, as democratic citizens, have to educate ourselves about tech and its so
cial and ethical implications — not only to decide what the regulations should be, but also to decide what role we
want big tech and social media to play in our lives,” said Sandel. Doing that will require a major educational int
ervention, both at Harvard and in higher education more broadly, he said. “We have to enable all students to learn
enough about tech and about the ethical implications of new technologies so that when they are running companies o
r when they are acting as democratic citizens, they will be able to ensure that technology serves human purposes r
ather than undermines a decent civic life.” Next: The AI revolution in medicine may lift personalized treatment, f
ill gaps in access to care, and cut red tape. Yet risks abound.

--- ARTICLE 4 ---
Title   : New AI tool can diagnose cancer, guide treatment, predict patient survival
Date   : 2024-09-04
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/2024/09/new-ai-tool-can-diagnose-cancer-guide-treatment-predict-p
atient-survival/

Content :
Video: Wirestock/Creatas Video+/Getty Images Plus Ekaterina Pesheva HMS Communications Scientists at Harvard Medic
al School have designed a versatile, ChatGPT-like AI model capable of performing an array of diagnostic tasks acro
ss multiple forms of cancers. The new AI system, described Wednesday in Nature, goes a step beyond many current A
I approaches to cancer diagnosis, the researchers said. Current AI systems are typically trained to perform specif
ic tasks — such as detecting cancer presence or predicting a tumor’s genetic profile — and they tend to work only
in a handful of cancer types. By contrast, the new model can perform a wide array of tasks and was tested on 19 ca
ncer types, giving it a flexibility like that of large language models such as ChatGPT. While other foundation AI
models for medical diagnosis based on pathology images have emerged recently, this is believed to be the first to
predict patient outcomes and validate them across several international patient groups. “Our ambition was to creat
e a nimble, versatile ChatGPT-like AI platform that can perform a broad range of cancer evaluation tasks,” said st
udy senior author Kun-Hsing Yu, assistant professor of biomedical informatics in the Blavatnik Institute at Harvar
d Medical School. “Our model turned out to be very useful across multiple tasks related to cancer detection, progn
osis, and treatment response across multiple cancers.” The team’s latest work builds on Yu’s previous research in
AI systems for the evaluation of colon cancer and brain tumors. These earlier studies demonstrated the feasibility
of the approach within specific cancer types and specific tasks. The new model, called CHIEF (Clinical Histopathol
ogy Imaging Evaluation Foundation), was trained on 15 million unlabeled images chunked into sections of interest.
The tool was then trained further on 60,000 whole-slide images of tissues including lung, breast, prostate, colore
ctal, stomach, esophageal, kidney, brain, liver, thyroid, pancreatic, cervical, uterine, ovarian, testicular, ski
n, soft tissue, adrenal gland, and bladder. Training the model to look both at specific sections of an image and t
he whole image allowed it to relate specific changes in one region to the overall context. This approach, the rese
archers said, enabled CHIEF to interpret an image more holistically by considering a broader context, instead of j
ust focusing on a particular region. CHIEF achieved nearly 94 percent accuracy in cancer detection and significant
ly outperformed current AI approaches across 15 datasets containing 11 cancer types. Following training, the team
tested CHIEF’s performance on more than 19,400 whole-slide images from 32 independent datasets collected from 24 h
ospitals and patient cohorts across the globe. The AI model, which works by reading digital slides of tumor tissue
s, detects cancer cells and predicts a tumor’s molecular profile based on cellular features seen on the image with
superior accuracy to most current AI systems. It can forecast patient survival across multiple cancer types and ac
curately pinpoint features in the tissue that surrounds a tumor — also known as the tumor microenvironment — that
are related to a patient’s response to standard treatments, including surgery, chemotherapy, radiation, and immuno
therapy. Finally, the team said, the tool appears capable of generating novel insights — it identified specific tu
mor characteristics previously not known to be linked to patient survival.

The findings, the research team said, add to growing evidence that AI-powered approaches can enhance clinicians’ a
bility to evaluate cancers efficiently and accurately, including the identification of patients who might not resp
ond well to standard cancer therapies.

“If validated further and deployed widely, our approach, and approaches similar to ours, could identify early on c
ancer patients who may benefit from experimental treatments targeting certain molecular variations, a capability t
hat is not uniformly available across the world,” Yu said. Overall, CHIEF outperformed other state-of-the-art AI m

ethods by up to 36 percent on the following tasks: cancer cell detection, tumor origin identification, predicting patient outcomes, and identifying the presence of genes and DNA patterns related to treatment response.Because of its versatile training, CHIEF performed equally well no matter how the tumor cells were obtained — whether via bio psy or through surgical excision. And it was just as accurate, regardless of the technique used to digitize the ca ncer cell samples. This adaptability, the researchers said, renders CHIEF usable across different clinical setting s and represents an important step beyond current models that tend to perform well only when reading tissues obtai ned through specific techniques. CHIEF achieved nearly 94 percent accuracy in cancer detection and significantly o utperformed current AI approaches across 15 datasets containing 11 cancer types. In five biopsy datasets collected from independent cohorts, CHIEF achieved 96 percent accuracy across multiple cancer types including esophagus, sto mach, colon, and prostate. When the researchers tested CHIEF on previously unseen slides from surgically removed t umors of the colon, lung, breast, endometrium, and cervix, the model performed with more than 90 percent accuracy. A tumor's genetic makeup holds critical clues to determine its future behavior and optimal treatments. To get this information, oncologists order DNA sequencing of tumor samples, but such detailed genomic profiling of cancer tiss ues is not done routinely nor uniformly across the world due to the cost and time involved in sending samples to s pecialized DNA sequencing labs. Even in well-resourced regions, the process could take several weeks. It's a gap t hat AI could fill, Yu said. Quickly identifying cellular patterns on an image suggestive of specific genomic aber rations could offer a quick and cost-effective alternative to genomic sequencing, the researchers said.

CHIEF outperformed current AI methods for predicting genomic variations in a tumor by looking at the microscopic s lides. This new AI approach successfully identified features associated with several important genes related to ca ncer growth and suppression, and it predicted key genetic mutations related to how well a tumor might respond to v arious standard therapies. CHIEF also detected specific DNA patterns related to how well a colon tumor might respo nd to a form of immunotherapy called immune checkpoint blockade. When looking at whole-tissue images, CHIEF ident ified mutations in 54 commonly mutated cancer genes with an overall accuracy of more than 70 percent, outperformin g the current state-of-the-art AI method for genomic cancer prediction. Its accuracy was greater for specific gene s in specific cancer types. The team also tested CHIEF on its ability to predict mutations linked with response to FDA-approved targeted therapies across 18 genes spanning 15 anatomic sites. CHIEF attained high accuracy in multip le cancer types, including 96 percent in detecting a mutation in a gene called EZH2 common in a blood cancer calle d diffuse large B-cell lymphoma. It achieved 89 percent for BRAF gene mutation in thyroid cancer, and 91 percent f or NTRK1 gene mutation in head and neck cancers. "Our ambition was to create a nimble, versatile ChatGPT-like AI platform that can perform a broad range of cancer evaluation tasks." CHIEF successfully predicted patient survival based on tumor histopathology images obtained at the time of initial diagnosis. In all cancer types and all patien t groups under study, CHIEF distinguished patients with longer-term survival from those with shorter-term surviva l. CHIEF outperformed other models by 8 percent. And in patients with more advanced cancers, CHIEF outperformed o ther AI models by 10 percent. In all, CHIEF's ability to predict high versus low death risk was tested and confirm ed across patient samples from 17 different institutions. The model identified tell-tale patterns on images relate d to tumor aggressiveness and patient survival. To visualize these areas of interest, CHIEF generated heat maps on an image. When human pathologists analyzed these AI-derived hot spots, they saw intriguing signals reflecting int eractions between cancer cells and surrounding tissues. One such feature was the presence of greater numbers of im mune cells in areas of the tumor in longer-term survivors, compared with shorter-term survivors. That finding, Yu noted, makes sense because a greater presence of immune cells may indicate the immune system has been activated to attack the tumor. When looking at the tumors of shorter-term survivors, CHIEF identified regions of interest marke d by the abnormal size ratios between various cell components, more atypical features on the nuclei of cells, weak connections between cells, and less presence of connective tissue in the area surrounding the tumor. These tumors also had a greater presence of dying cells around them. For example, in breast tumors, CHIEF pinpointed as an are a of interest the presence of necrosis — or cell death — inside the tissues. On the flip side, breast cancers with higher survival rates were more likely to have preserved cellular architecture resembling heathy tissues. The visu al features and zones of interest related to survival varied by cancer type, the team noted. The researchers said they plan to refine CHIEF's performance and augment its capabilities by: Authorship: Co-authors included Xiyue Wan g, Junhan Zhao, Eliana Marostica, Wei Yuan, Jietian Jin, Jiayu Zhang, Ruijiang Li, Hongping Tang, Kanran Wang, Yu Li, Fang Wang, Yulong Peng, Junyou Zhu, Jing Zhang, Christopher R. Jackson, Jun Zhang, Deborah Dillon, Nancy U. Li n, Lynette Sholl, Thomas Denize, David Meredith, Keith L. Ligon, Sabina Signoretti, Shuji Ogino, Jeffrey A. Golde n, MacLean P. Nasrallah, Xiao Han, Sen Yang. Disclosures: Yu is an inventor of U.S. patent 16/179,101 assigned to Harvard University and served as a consultant for Takeda, Curatio DL, and the Postgraduate Institute for Medicine. Jun Zhang and Han were employees of Tencent AI Lab.

The work was in part supported by the National Institute of General Medical Sciences grant R35GM142879, the Depart ment of Defense Peer Reviewed Cancer Research Program Career Development Award HT9425-23-1-0523, the Google Resear ch Scholar Award, the Harvard Medical School Dean's Innovation Award, and the Blavatnik Center for Computational B iomedicine Award.

--- ARTICLE 5 ---
Title   : AI revolution in medicine
Date   : 2020-11-11
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/2020/11/risks-and-benefits-of-an-ai-revolution-in-medicine/

Content :
Illustration by Ben Boothman Alvin Powell Harvard Staff Writer Third in a series that taps the expertise of the Ha rvard community to examine the promise and potential pitfalls of the coming age of artificial intelligence and mac hine learning. The news is bad: "I'm sorry, but you have cancer." Those unwelcome words sink in for a few minutes, and then your doctor begins describing recent advances in artificial intelligence, advances that let her compare y our case to the cases of every other patient who's ever had the same kind of cancer. She says she's found the most

effective treatment, one best suited for the specific genetic subtype of the disease in someone with your genetic background — truly personalized medicine. And the prognosis is good. It has taken time — some say far too long — but medicine stands on the brink of an AI revolution. In a recent article in the New England Journal of Medicine, Isaac Kohane, head of Harvard Medical School’s Department of Biomedical Informatics, and his co-authors say that AI will indeed make it possible to bring all medical knowledge to bear in service of any case. Properly designed AI also has the potential to make our health care system more efficient and less expensive, ease the paperwork burden that has more and more doctors considering new careers, fill the gaping holes in access to quality care in the world’s poorest places, and, among many other things, serve as an unblinking watchdog on the lookout for the medical errors that kill an estimated 200,000 people and cost $1.9 billion annually. “I’m convinced that the implementation of AI in medicine will be one of the things that change the way care is delivered going forward,” said David Bates, chief of internal medicine at Harvard-affiliated Brigham and Women’s Hospital, professor of medicine at Harvard Medical School and of health policy and management at the Harvard T.H. Chan School of Public Health. “It’s clear that clinicians don’t make as good decisions as they could. If they had support to make better decisions, they could do a better job.” Years after AI permeated other aspects of society, powering everything from creepily sticky online ads to financial trading systems to kids’ social media apps to our increasingly autonomous cars, the proliferation of studies showing the technology’s algorithms matching the skill of human doctors at a number of tasks signals its imminent arrival. “I think it’s an unstoppable train in a specific area of medicine — showing true expert-level performance — and that’s in image recognition,” said Kohane, who is also the Marion V. Nelson Professor of Biomedical Informatics. “Once again medicine is slow to the mark. I’m no longer irritated but bemused that my kids, in their social sphere, are using more advanced AI than I use in my practice.” But even those who see AI’s potential value recognize its potential risks. Poorly designed systems can misdiagnose. Software trained on data sets that reflect cultural biases will incorporate those blind spots. AI designed to both heal and make a buck might increase — rather than cut — costs, and programs that learn as they go can produce a raft of unintended consequences once they start interacting with unpredictable humans. “I think the potential of AI and the challenges of AI are equally big,” said Ashish Jha, former director of the Harvard Global Health Institute and now dean of Brown University’s School of Public Health. “There are some very large problems in health care and medicine, both in the U.S. and globally, where AI can be extremely helpful. But the costs of doing it wrong are every bit as important as its potential benefits. … The question is: Will we be better off?” Many believe we will, but caution that implementation has to be done thoughtfully, with recognition of not just AI’s strengths but also its weaknesses, and taking advantage of a range of viewpoints brought by experts in fields outside of medicine and computer science, including ethics and philosophy, sociology, psychology, behavioral economics, and, one day, those trained in the budding field of machine behavior, which seeks to understand the complex and evolving interaction of humans and machines that learn as they go. — Finale Doshi-Velez, John L. Loeb Associate Professor of Engineering and Applied Sciences at the Harvard John A. Paulson School of Engineering and Applied Sciences Rose Lincoln/Harvard file photo “The challenge with machine behavior is that you’re not deploying an algorithm in a vacuum. You’re deploying it into an environment where people will respond to it, will adapt to it. If I design a scoring system to rank hospitals, hospitals will change,” said David Parkes, George F. Colony Professor of Computer Science, co-director of the Harvard Data Science Initiative, and one of the co-authors of a recent article in the journal Nature calling for the establishment of machine behavior as a new field. “Just as it would be challenging to understand how a new employee will do in a new work environment, it’s challenging to understand how machines will do in any kind of environment, because people will adapt to them, will change their behavior.” Though excitement has been building about the latest wave of AI, the technology has been in medicine for decades in some form, Parkes said. As early as the 1970s, “expert systems” were developed that encoded knowledge in a variety of fields in order to make recommendations on appropriate actions in particular circumstances. Among them was Mycin, developed by Stanford University researchers to help doctors better diagnose and treat bacterial infections. Though Mycin was as good as human experts at this narrow chore, rule-based systems proved brittle, hard to maintain, and too costly, Parkes said. The excitement over AI these days isn’t because the concept is new. It’s owing to rapid progress in a branch called machine learning, which takes advantage of recent advances in computer processing power and in big data that have made compiling and handling massive data sets routine. Machine learning algorithms — sets of instructions for how a program operates — have become sophisticated enough that they can learn as they go, improving performance without human intervention. “The superpower of these AI systems is that they can look at all of these large amounts of data and hopefully surface the right information or the right predictions at the right time,” said Finale Doshi-Velez, John L. Loeb Associate Professor of Engineering and Applied Sciences at the Harvard John A. Paulson School of Engineering and Applied Sciences (SEAS). “Clinicians regularly miss various bits of information that might be relevant in the patient’s history. So that’s an example of a relatively low-hanging fruit that could potentially be very useful.” Before being used, however, the algorithm has to be trained using a known data set. In medical imaging, a field where experts say AI holds the most promise soonest, the process begins with a review of thousands of images — of potential lung cancer, for example — that have been viewed and coded by experts. Using that feedback, the algorithm analyzes an image, checks the answer, and moves on, developing its own expertise. In recent years, increasing numbers of studies show machine-learning algorithms equal and, in some cases, surpass human experts in performance. In 2016, for example, researchers at Beth Israel Deaconess Medical Center reported that an AI-powered diagnostic program correctly identified cancer in pathology slides 92 percent of the time, just shy of trained pathologists’ 96 percent. Combining the two methods led to 99.5 percent accuracy. More recently, in December 2018, researchers at Massachusetts General Hospital (MGH) and Harvard’s SEAS reported a system that was as accurate as trained radiologists at diagnosing intracranial hemorrhages, which lead to strokes. And in May 2019, researchers at Google and several academic medical centers reported an AI designed to detect lung cancer that was 94 percent accurate, beating six radiologists and recording both fewer false positives and false negatives. — David Parkes, George F. Colony Professor of Computer Science and co-director of the Harvard Data Science Initiative Kris Snibbe/Harvard file photo One recent area where AI’s promise has remained largely unrealized is the global response to COVID-19, according to Kohane and Bates. Bates, who delivered a talk in August at the Riyad Global Digital Health Summit titled “Use of AI in Weathering the COVID Storm,” said though there were successes, much of the response has relied on traditional epidemiological and medical tools. One striking exception, he said, was the early detection of unusual pneumonia cases around a market in Wuhan, China, in late December by an AI system developed by Canada-based BlueDot. The detection, which would turn out to be SARS-CoV-2, came more than a week before the World Health Organization issued a public notice of the new virus. “We did some things with artificial intelligence in this pandemic, but there is much more that w e could do,” Bates told the online audience. In comments in July at the online conference FutureMed, Kohane was more succinct: “It was a very, very unimpressive performance. … We in health care were shooting for the moon, but we actually had not gotten out of our own backyard.” The two agree that the biggest impediment to greater use of AI in formulating COVID response has been a lack of reliable, real-time data. Data collection and sharing have been slowed by older infrastructure — some U.S. reports are still faxed to public health centers, Bates said — by lags in data collection, and by privacy concerns that short-circuit data sharing. “COVID has shown us that we have a data-access problem at the national and international level that prevents us from addressing burning problems in national health emergencies,” Kohane said. A key success, Kohane said, may yet turn out to be the use of machine learning in vaccine development. We won’t likely know for some months which candidates proved most successful, but Kohane pointed out that the technology was used to screen large databases and select which viral proteins offered the greatest chance of success if blocked by a vaccine. “It will play a much more important role going forward,” Bates said, expressing confidence that the current hurdles would be overcome. “It will be a key enabler of better management in the next pandemic.” Corporations agree about that future promise and in recent years have been scrambling to join in. In February 2019, IBM Watson Health began a 10-year, $50 million partnership with Brigham and Women’s Hospital and Vanderbilt University Medical Center whose aim is to use AI on electronic health records and claims data to improve patient safety, precision medicine, and health equity. And in March 2019, Amazon awarded a $2 million AI research grant to Beth Israel in an effort to improve hospital efficiency, including patient care and clinical workflows. A properly developed and deployed AI, experts say, will be akin to the cavalry riding in to help beleaguered physicians struggling with unrelenting workloads, high administrative burdens, and a tsunami of new clinical data. Robert Truog, head of the HMS Center for Bioethics, the Frances Glessner Lee Professor of Legal Medicine, and a pediatric anesthesiologist at Boston Children’s Hospital, said the defining characteristic of his last decade in practice has been a rapid increase in information. While more data about patients and their conditions might be viewed as a good thing, it’s only good if it can be usefully managed. — Robert Truog, head of the Harvard Medical School Center for Bioethics and the the Frances Glessner Lee Professor of Legal Medicine Rose Lincoln/Harvard file photo “Over the last 10 years of my career the volume of data has absolutely gone exponential,” Truog said. “I would have one image on a patient per day: their morning X-ray. Now, if you get an MRI, it generates literally hundreds of images, using different kinds of filters, different techniques, all of which convey slightly different variations of information. It’s just impossible to even look at all of the images. “Psychologists say that humans can handle four independent variables and when we get to five, we’re lost,” he said. “So AI is coming at the perfect time. It has the potential to rescue us from data overload.” Given the technology’s facility with medical imaging analysis, Truog, Kohane, and others say AI’s most immediate impact will be in radiology and pathology, fields where those skills are paramount. And, though some see a future with fewer radiologists and pathologists, others disagree. The best way to think about the technology’s future in medicine, they say, is not as a replacement for physicians, but rather as a force-multiplier and a technological backstop that not only eases the burden on personnel at all levels, but makes them better. “You’re not expecting this AI doctor that’s going to cure all ills but rather AI that provides support so better decisions can be made,” Doshi-Velez said. “Health is a very holistic space, and I don’t see AIs being anywhere near able to manage a patient’s health. It’s too complicated. There are too many factors, and there are too many factors that aren’t really recorded.” In a September 2019 issue of the Annals of Surgery, Ozanan Meireles, director of MGH’s Surgical Artificial Intelligence and Innovation Laboratory, and general surgery resident Daniel Hashimoto offered a view of what such a backstop might look like. They described a system that they’re training to assist surgeons during stomach surgery by having it view thousands of videos of the procedure. Their goal is to produce a system that one day could virtually peer over a surgeon’s shoulder and offer advice in real time. At the Harvard Chan School, meanwhile, a group of faculty members, including James Robins, Miguel Hernan, Sonia Hernandez-Diaz, and Andrew Beam, are harnessing machine learning to identify new interventions that can improve health outcomes. Their work, in the field of “causal inference,” seeks to identify different sources of the statistical associations that are routinely found in the observational studies common in public health. Those studies are good at identifying factors that are linked to each other but less able to identify cause and effect. Hernandez-Diaz, a professor of epidemiology and co-director of the Chan School’s pharmacoepidemiology program, said causal inference can help interpret associations and recommend interventions. For example, elevated enzyme levels in the blood can predict a heart attack, but lowering them will neither prevent nor treat the attack. A better understanding of causal relationships — and devising algorithms to sift through reams of data to find them — will let researchers obtain valid evidence that could lead to new treatments for a host of conditions. “We will make mistakes, but the momentum won’t go back the other way,” Hernandez-Diaz said of AI’s increasing presence in medicine. “We will learn from them.” Finding new interventions is one thing; designing them so health professionals can use them is another. Doshi-Velez’s work centers on “interpretable AI” and optimizing how doctors and patients can put it to work to improve health. AI’s strong suit is what Doshi-Velez describes as “large, shallow data” while doctors’ expertise is the deep sense they may have of the actual patient. Together, the two make a potentially powerful combination, but one whose promise will go unrealized if the physician ignores AI’s input because it is rendered in hard-to-use or unintelligible form. “I’m very excited about this team aspect and really thinking about the things that AI and machine-learning tools can provide an ultimate decision-maker — we’ve focused on doctors so far, but it could also be the patient — to empower them to make better decisions,” Doshi-Velez said. — Isaac Kohane, head of Harvard Medical School’s Department of Biomedical Informatics Stephanie Mitchell/Harvard file photo While many point to AI’s potential to make the health care system work better, some say its potential to fill gaps in medical resources is also considerable. In regions far from major urban medical centers, local physicians could be able to get assistance diagnosing and treating unfamiliar conditions and have available an AI-driven consultant that allows them to offer patients a specialists’ insight as they decide whether a particular procedure — or additional expertise — is needed. Outside the developed world that capability has the potential to be transformative, according to Jha. AI-powered applications have the potential to vastly improve care in places where doctors are absent, and informal medical systems have risen to fill the need. Recent studies in India and China serve as powerful examples. In India’s Bihar state, for example, 86 percent of cases resulted in unneeded or harmful medicine being prescribed. Even in urban Delhi, 54 percent of cases resulted in unneeded or harmful medicine. “If you are sick, is it better to go to the doctor or not? In 2019, in large parts of the world, it’s a wash. It’s unclear. And that is scary,” Jha said. “So it’s a low bar. People ask, ‘Will AI be helpful?’ I say we’d really have to screw up AI for it not to be helpful. Net-net, the opportunity for improvement over the status quo is massive.” Though the promise is great, the road ahead isn’t necessarily smooth. Even AI’s most ardent supporters acknowledge that the likely bumps and potholes, both seen and unseen, should be taken seriously. One challenge is ensuring that high-quality data is used to train AI. If it is biased or otherwise flawed, that will be reflected in the performance. A second challenge is ensuring that the prejudices rife in society aren't reflected in the algorithms, added by programmers unaware of those they may unconsciously hold. That potential was a central point in a 2016 Wisconsin legal case, when an AI-driven, risk-assessment system for criminal recidivism was used in sentencing a man to six years in prison. The judge remarked that the "risk-assessment tools that have been utilized suggest that you're extremely high risk to reoffend." The defendant challenged the sentence, arguing that the AI's proprietary software — which he couldn't examine — may have violated his right to be sentenced based on accurate information. The sentence was upheld by the state supreme court, but that case, and the spread of similar systems to assess pretrial risk, has generated national debate over the potential for injustices due to our increasing reliance on systems that have power over freedom or, in the health care arena, life and death, and that may be unfairly tilted or outright wrong. "We have to recognize that getting diversity in the training of these algorithms is going to be incredibly important, otherwise we will be in some sense pouring concrete over whatever current distortions exist in practice, such as those due to socioeconomic status, ethnicity, and so on," Kohane said. Also highlighted by the case is the "black box" problem. Since the algorithms are designed to learn and improve their performance over time, sometimes even their designers can't be sure how they arrive at a recommendation or diagnosis, a feature that leaves some uncomfortable. — Ashish Jha, former director of the Harvard Global Health Institute and now dean of Brown University's School of Public Health Jon Chase/Harvard Staff Photographer "If you start applying it, and it's wrong, and we have no ability to see that it's wrong and to fix it, you can cause more harm than good," Jha said. "The more confident we get in technology, the more important it is to understand when humans can override these things. I think the Boeing 737 Max example is a classic example. The system said the plane is going up, and the pilots saw it was going down but couldn't override it." Jha said a similar scenario could play out in the developing world should, for example, a community health worker see something that makes him or her disagree with a recommendation made by a big-name company's AI-driven app. In such a situation, being able to understand how the app's decision was made and how to override it is essential. "If you see a frontline community health worker in India disagree with a tool developed by a big company in Silicon Valley, Silicon Valley is going to win," Jha said. "And that's potentially a dangerous thing." Researchers at SEAS and MGH's Radiology Laboratory of Medical Imaging and Computation are at work on the two problems. The AI-based diagnostic system to detect intracranial hemorrhages unveiled in December 2019 was designed to be trained on hundreds, rather than thousands, of CT scans. The more manageable number makes it easier to ensure the data is of high quality, according to Hyunkwang Lee, a SEAS doctoral student who worked on the project with colleagues including Sehyo Yune, a former postdoctoral research fellow at MGH Radiology and co-first author of a paper on the work, and Synho Do, senior author, HMS assistant professor of radiology, and director of the lab. "We ensured the data set is of high quality, enabling the AI system to achieve a performance similar to that of radiologists," Lee said. Second, Lee and colleagues figured out a way to provide a window into an AI's decision-making, cracking open the black box. The system was designed to show a set of reference images most similar to the CT scan it analyzed, allowing a human doctor to review and check the reasoning. Jonathan Zittrain, Harvard's George Bemis Professor of Law and director of the Berkman Klein Center for Internet and Society, said that, done wrong, AI in health care could be analogous to the cancer-causing asbestos that was used for decades in buildings across the U.S., with widespread harmful effects not immediately apparent. Zittrain pointed out that image analysis software, while potentially useful in medicine, is also easily fooled. By changing a few pixels of an image of a cat — still clearly a cat to human eyes — MIT students prompted Google image software to identify it, with 100 percent certainty, as guacamole. Further, a well-known study by researchers at MIT and Stanford showed that three commercial facial-recognition programs had both gender and skin-type biases. Ezekiel Emanuel, a professor of medical ethics and health policy at the University of Pennsylvania's Perelman School of Medicine and author of a recent Viewpoint article in the Journal of the American Medical Association, argued that those anticipating an AI-driven health care transformation are likely to be disappointed. Though he acknowledged that AI will likely be a useful tool, he said it won't address the biggest problem: human behavior. Though they know better, people fail to exercise and eat right, and continue to smoke and drink too much. Behavior issues also apply to those working within the health care system, where mistakes are routine. "We need fundamental behavior change on the part of these people. That's why everyone is frustrated: Behavior change is hard," Emanuel said. Susan Murphy, professor of statistics and of computer science, agrees and is trying to do something about it. She's focusing her efforts on AI-driven mobile apps with the aim of reinforcing healthy behaviors for people who are recovering from addiction or dealing with weight issues, diabetes, smoking, or high blood pressure, conditions for which the personal challenge persists day by day, hour by hour. The sensors included in ordinary smartphones, augmented by data from personal fitness devices such as the ubiquitous Fitbit, have the potential to give a well-designed algorithm ample information to take on the role of a health care angel on your shoulder. The tricky part, Murphy said, is to truly personalize the reminders. A big part of that, she said, is understanding how and when to nudge — not during a meeting, for example, or when you're driving a car, or even when you're already exercising, so as to best support adopting healthy behaviors. "How can we provide support for you in a way that doesn't bother you so much that you're not open to help in the future?" Murphy said. "What our algorithms do is they watch how responsive you are to a suggestion. If there's a reduction in responsivity, they back off and come back later." The apps can use sensors on your smartphone to figure out what's going on around you. An app may know you're in a meeting from your calendar, or talking more informally from ambient noise its microphone detects. It can tell from the phone's GPS how far you are from a gym or an AA meeting or whether you are driving and so should be left alone. Trickier still, Murphy said, is how to handle moments when the AI knows more about you than you do. Heart rate sensors and a phone's microphone might tell an AI that you're stressed out when your goal is to live more calmly. You, however, are focused on an argument you're having, not its physiological effects and your long-term goals. Does the app send a nudge, given that it's equally possible that you would take a calming breath or angrily toss your phone across the room? Working out such details is difficult, albeit key, Murphy said, in order to design algorithms that are truly helpful, that know you well, but are only as intrusive as is welcome, and that, in the end, help you achieve your goals. For AI to achieve its promise in health care, algorithms and their designers have to understand the potential pitfalls. To avoid them, Kohane said it's critical that AIs are tested under real-world circumstances before wide release. Similarly, Jha said it's important that such systems aren't just released and forgotten. They should be reevaluated periodically to ensure they're functioning as expected, which would allow for faulty AIs to be fixed or halted altogether. Several experts said that drawing from other disciplines — in particular ethics and philosophy — may also help. Programs like Embedded EthiCS at SEAS a nd the Harvard Philosophy Department, which provides ethics training to the University's computer science students, seek to provide those who will write tomorrow's algorithms with an ethical and philosophical foundation that will help them recognize bias — in society and themselves — and teach them how to avoid it in their work. Disciplines dealing with human behavior — sociology, psychology, behavioral economics — not to mention experts on policy, government regulation, and computer security, may also offer important insights. "The place we're likely to fall down is the way in which recommendations are delivered," Bates said. "If they're not delivered in a robust way, providers will ignore them. It's very important to work with human factor specialists and systems engineers about the way that suggestions are made to patients." Bringing these fields together to better understand how AIs work once they're "in the wild" is the mission of what Parkes sees as a new discipline of machine behavior. Computer scientists and health care experts should seek lessons from sociologists, psychologists, and cognitive behaviorists in answering questions about whether an AI-driven system is working as planned, he said. "How useful was it that the AI system proposed that this medical expert should talk to this other medical expert?" Parkes said. "Was that intervention followed? Was it a productive conversation? Would they have talked anyway? Is there any way to tell?" Next: A Harvard project asks people to envision how technology will change their lives going forward.

--- ARTICLE 6 ---
Title   : Using AI to repurpose existing drugs for treatment of rare diseases
Date   : 2024-09-25
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/2024/09/using-ai-to-repurpose-existing-drugs-for-treatment-of-rare-diseases/

Content :
Identifies possible therapies for thousands of diseases, including ones with no current treatments Ekaterina Pesheva HMS Communications There are more than 7,000 rare and undiagnosed diseases globally. Although each condition occurs in a small number of individuals, collectively these diseases exert a staggering human and economic toll because they affect some 300 million people worldwide. Yet, with a mere 5 to 7 percent of these conditions having an FDA-approved drug, they remain largely untreated or undertreated. Developing new medicines represents a daunting challenge, but a new artificial intelligence tool can propel the discovery of new therapies from existing medicines, offering hope for patients with rare and neglected conditions and for the clinicians who treat them. The AI model, called TxGNN, is the first one developed specifically to identify drug candidates for rare diseases and conditions with no treatments. It identified drug candidates from existing medicines for more than 17,000 diseases, many of them without any existing treatments. This represents the largest number of diseases that any single AI model can handle to date. The researchers note that the model could be applied to even more diseases beyond the 17,000 it worked on in the initial experiments. The work, described Wednesday in Nature Medicine, was led by scientists at Harvard Medical School. The researchers have made the tool available for free and want to encourage clinician-scientists to use it in their search for new therapies, especially for conditions with no or with limited treatment options.   "With this tool we aim to identify new therapies across the disease spectrum but when it comes to rare, ultrarare, and neglected conditions, we foresee this model could help close, or at least narrow, a gap that creates serious health disparities," said lead researcher Marinka Zitnik, assistant professor of biomedical informatics in the Blavatnik Institute at HMS. "This is precisely where we see the promise of AI in reducing the global disease burden, in finding new uses for existing drugs, which is also a faster and more cost-effective way to develop therapies than designing new drugs from scratch," added Zitnik, who is an associate faculty member at the Kempner Institute for the Study of Natural and Artificial Intelligence at Harvard University. "When it comes to rare, ultrarare, and neglected conditions, we foresee this model could help close, or at least narrow, a gap that creates serious health disparities." The new tool has two central features — one that identifies treatment candidates along with possible side effects and another one that explains the rationale for the decision.   In total, the tool identified drug candidates from nearly 8,000 medicines (both FDA-approved medicines and experimental ones now in clinical trials) for 17,080 diseases, including conditions with no available treatments. It also predicted which drugs would have side effects and contraindications for specific conditions — something that the current drug discovery approach identifies mostly by trial and error during early clinical trials focused on safety. Compared against the leading AI models for drug repurposing, the new tool was nearly 50 percent better, on average, at identifying drug candidates. It was also 35 percent more accurate in predicting what drugs would have contraindications. Repurposing existing drugs is an alluring way to develop new treatments because it relies on medicines that have been studied, have well-understood safety profiles, and have gone through the regulatory approval process. Most medicines have multiple effects beyond the specific targets they were originally developed and approved for. But many of these effects remain undiscovered and understudied during initial testing, clinical trials, and review, only emerging after years of use by millions of people. Indeed, nearly 30 percent of FDA- approved drugs have acquired at least one additional indication for treatment following initial approval, and many have acquired tens of additional treatment indications over the years. This approach to drug repurposing is haphazard at best. It relies on patient reports of unexpected beneficial side effects or on physicians' intuition about whether to use a drug for a condition that it was not intended for, a practice known as off-label use. "We've tended to rely on luck and serendipity rather than on strategy, which limits drug discovery to diseases for which drugs already exist," Zitnik said. The benefits of drug repurposing extend beyond diseases without treatments, Zitnik noted. "Even for more common diseases with approved treatments, new drugs could offer alternatives with fewer side effects or replace drugs that are ineffective for certain patients," she said. Most current AI models used for drug discovery are trained on a single disease or a handful of conditions. Rather than focusing on specific diseases, the new tool was trained in a manner that enables it to use existing data to make new predictions. It does so by identifying shared features across multiple diseases, such as shared genomic aberrations. For example, the AI model pinpoints shared disease mechanisms based on common genomic underpinnings, which allows it to extrapolate from a well-understood disease with known treatments to a poorly understood one with no treatments. This capacity, the research team said, brings the AI tool closer to the type of a reasoning a human clinician might use to generate novel ideas if they had access to all the preexisting knowledge and raw data that the AI model does but that the human brain cannot possibly access or sto

re. The tool was trained on vast amounts of data, including DNA information, cell signaling, levels of gene activity, clinical notes, and more. The researchers tested and refined the model by asking it to perform various tasks. Finally, the tool’s performance was validated on 1.2 million patient records and asked to identify drug candidates for various diseases. The researchers also asked the tool to predict what patient characteristics would render the identified drug candidates contraindicated for certain patient populations. Another task involved asking the tool to identify existing small molecules that might effectively block the activity of certain proteins implicated in disease-causing pathways and processes. In a test designed to gauge the model’s ability to reason as a human clinician might, the researchers prompted the model to find drugs for three rare conditions it had not seen as part of its training — a neurodevelopmental disorder, a connective-tissue disease, and a rare genetic condition that causes water imbalance. The researchers then compared the model’s recommendations for drug therapy against current medical knowledge about how the suggested drugs work. In every example, the tool’s recommendations aligned with current medical knowledge. Moreover, the model not only identified medicines for all three diseases but also provided the rationale behind its decision. This explainer feature allows for transparency and can increase physician confidence.

The researchers caution that any therapies identified by the model would require additional evaluation for dosing and timing of delivery. But, they add, with this unprecedented capacity, the new AI model would expedite drug repurposing in a manner not possible until now. The team is already collaborating with several rare disease foundations to help identify possible treatments. Co-authors included Kexin Huang, Payal Chandak, Qianwen Wang, Shreyas Havaldar, Akhil Vaid, Jure Leskovec, Girish N. Nadkarni, Benjamin S. Glicksberg, and Nils Gehlenborg. This work was supported by National Science Foundation CAREER award (grant 2339524), National Institutes of Health (grant R01-HD108794), U.S. Department of Defense (grant FA8702-15-D-0001), Amazon Faculty Research, Google Research Scholar Program, AstraZeneca Research, Roche Alliance with Distinguished Scientists, Sanofi iDEA-TECH Award, Pfizer Research, Chan Zuckerberg Initiative, John and Virginia Kaneb Fellowship at HMS, Biswas Family Foundation Transformative Computational Biology Grant in partnership with the Milken Institute, HMS Dean’s Innovation Awards for the Use of Artificial Intelligence, Kempner Institute for the Study of Natural and Artificial Intelligence at Harvard University, and Dr. Susanne E. Churchill Summer Institute in Biomedical Informatics at HMS.

--- ARTICLE 7 ---

Title   : It’s inoperable cancer. Should AI make call about what happens next?
Date   : 2025-02-10
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/2025/02/should-ai-be-used-in-end-of-life-medical-decisions-a-i-artificial-intelligence/

Content :

Rebecca Weintraub Brendel, director of Harvard Medical School’s Center for Bioethics. Veasey Conway/Harvard Staff Photographer Arrival of large-language models sparking discussion of how use of technology may be broadened in patient care, and what it means to be human Alvin Powell Harvard Staff Writer AI is already being used in clinics to help analyze imaging data, such as X-rays and scans. But the recent arrival of sophisticated large-language AI models on the scene is forcing consideration of broadening the use of the technology into other areas of patient care. In this edited conversation with the Gazette, Rebecca Weintraub Brendel, director of Harvard Medical School’s Center for Bioethics, looks at end-of-life options and the importance of remembering that just because we can, doesn’t always mean we should. When we talk about artificial intelligence and end-of-life decision-making, what are the important questions at play? End-of-life decision-making is the same as other decision-making because ultimately, we do what patients want us to do, provided they are competent to make those decisions and what they want is medically indicated — or at least not medically contraindicated. One complication would be if a patient is so ill that they can’t tell us what they want. The second challenge is understanding in both a cognitive way and an emotional way what the decision means. People sometimes say, “I would never want to live that way” but they wouldn’t make the same decision in all circumstances. Patients who’ve lived with progressive neurologic conditions like ALS for a long time often have a sense of when they’ve reached their limit. They’re not depressed or frightened and are ready to make their decision. On the other hand, depression is quite prevalent in some cancers and people tend to change their minds about wanting to end their lives once symptoms are treated. So, if someone is young and says, “If I lose my legs, I wouldn’t want to live,” should we allow for shifting perspectives as we get to the end of life? When we’re faced with something that alters our sense of bodily integrity, our sense of ourselves as fully functional human beings, it’s natural, even expected, that our capacity to cope can be overwhelmed. But there are pretty devastating injuries where a year later, people report having a better quality of life than before, even for severe spinal cord injuries and quadriplegia. So, we can overcome a lot, and our capacity for change, for hope, has to be taken into account. So, how do we, as healers of mind and body, help patients make decisions about their end of life? “I’d be hard-pressed to say that we’d ever want to give away our humanity in making decisions of high consequence.” For someone with a chronic illness, the standard of care has those decisions happening along the way, and AI could be helpful there. But at the point of diagnosis — do I want treatment or to opt for palliation from the beginning — AI might give us a sense of what one might anticipate, how impaired we might be, whether pain can be palliated, or what the tipping point will be for an individual person. So, the ability to have AI gather and process orders of magnitude more information than what the human mind can process — without being colored by fear, anxiety, responsibility, relational commitments — might give us a picture that could be helpful. What about the patient who is incapacitated, with no family, no advance directives, so the decision falls to the care team? We have to have an attitude of humility toward these decisions. Having information can be really helpful. With somebody who’s never going to regain capacity, we’re stuck with a few different options. If we really don’t know what they would like, because they’re somebody who avoided treatment and really didn’t want to be in the hospital, or didn’t have a lot of relationships, we assume that they wouldn’t have sought treatment for something that was life-ending. But we have to be aware that we’re making a lot of assumptions, even if we’re not necessarily doing the wrong thing. Having a better prognostic sense of what might happen is really important to that decision, which, again, is where A

I can help. I’m less optimistic about the use of large-language models for making capacity decisions or figuring o
ut what somebody would have wanted. To me it’s about respect. We respect our patients and try to make our best gue
sses, and realize that we all are complicated, sometimes tortured, sometimes lovable, and, ideally, loved. Are the
re things that AI should not be allowed to do? I’m sure it could make end-of-life recommendations versus simply ga
thering information. We have to be careful where we use “is” to make an “ought” decision. If AI told you that ther
e is less than 5 percent chance of survival, that alone is not enough to tell us what we ought to do. If there’s b
een a terrible tragedy or a violent assault on someone, we would look at that 5 percent differently from someone w
ho’s been battling a chronic illness over time and says, “I don’t want to go through this again, and I don’t want
to put others through this. I’ve had a wonderful life.” “If AI told you that there is less than 5 percent chance o
f survival, that alone is not enough to tell us what we ought to do.” In diagnostic and prognostic assessments, AI
has already started to outperform physicians, but that doesn’t answer the critical question of how we interpret th
at, in terms of what our default rules should be about human behavior. It can help us be more transparent and acco
untable and respectful of each other by making it explicit that, as a society, if these things happen, unless you
tell us otherwise we’re not going to resuscitate. Or we are when we think there’s a good chance of recovery. I do
n’t want to underestimate AI’s potential impact, but we can’t abdicate our responsibility to center human meaning
in our decisions, even when based on data. So these decisions should always be made by humans? “Always” is a reall
y strong word, but I’d be hard-pressed to say that we’d ever want to give away our humanity in making decisions of
high consequence. Are there areas of medicine where people should always be involved? Should a baby’s first contac
t with the world always be human hands? Or should we just focus on quality of care? I would want people around, ev
en if a robot does the surgery because the outcome is better. We would want to maintain the human meaning of impor
tant life events. Another question that comes up is, what will it mean to be a physician, a healer, a healthcare p
rofessional? We hold a lot of information and an information asymmetry is one of the things that has caused medica
l and other healthcare professionals to be held in high esteem. But it’s also about what we do with the informatio
n, being a great diagnostician, having an exemplary bedside manner, and ministering to patients at a time when the
y’re suffering. How do we redefine the profession when the things we thought we were best at, we may not be the be
st at anymore? At some point, we may have to question human interaction in the system. Does it introduce bias, and
to what extent is processing by human minds important? Is an LLM going to create new information, come up with a n
ew diagnostic category, or a disease entity? What ought the responsibilities of patients and doctors be to each ot
her in a hyper-technological age? Those are important questions that we need to look at. Are those conversations h
appening? Yes. In our Center for Bioethics, one of the things that we’re looking at is how does artificial intelli
gence look at some of our timeless challenges within health? Technology tends to go where there’s capital and reso
urces, while LLMs and AI advances could allow us to care for swaths of the population where there’s no doctor with
in a day’s travel. Holding ourselves accountable on questions of equity, justice, and advancing global health is r
eally important. There are questions about moral leadership in medicine. How do we make sure that output from LLMs
and future iterations of AIs comport with the people we think we are and the people we ought to be? How should we
educate to make sure that the values of the healing professions continue to be front and center in delivering car
e? How do we balance the public’s health and individual health, and how does that play out in other countries? So,
when we talk about patients in under-resourced settings and about AI’s capabilities versus what it means to be hum
an, we need to be mindful that in some parts of the world to be human is to suffer and not have access to care? Ye
s, because, increasingly, we can do something about it. As we’re developing tools that can allow us to make huge d
ifferences in practical and affordable ways, we have to ask, “How do we do that and follow our values of justice,
care, respect for persons? How do we make sure that we don’t abandon them when we actually have the capacity to he
lp?”

--- ARTICLE 8 ---
Title   : AI may be just what the dentist ordered
Date   : 2023-11-29
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/2023/11/harvard-symposium-looks-at-ai-and-oral-health/

Content :
Kat J. McAlpine Harvard Correspondent Dental practices, dental schools, oral health researchers, and policymakers
are rapidly positioning themselves to evolve in step with the dawning AI movement in oral healthcare, say AI and d
entistry experts, who shared their insights at the inaugural Global Symposium on AI & Dentistry at Harvard, Nov. 3
-4. “AI holds the promise of transforming the way we practice oral healthcare, pinpoint and treat diseases and con
ditions, and increase equitable access to care and treatment,” said William Giannobile, dean of Harvard School of
Dental Medicine, during his opening remarks. Three hundred attendees from 30 countries joined the symposium in per
son, with another 120 tuning in virtually. More than 65 research projects were presented, featuring a range of dev
ice prototypes, patient-facing smartphone apps, and other technologies under development at the intersection of AI
and dentistry. For more than 40 years, researchers have been experimenting with ways to apply AI to dentistry, sai
d Florian Hillen, founder and CEO of VideaHealth, a dental imaging startup launched from AI research conducted at
Harvard and MIT. Within the last decade, AI capabilities have finally reached critical mass. “AI-powered tools are
now helping dentists identify dental decay in patients up to five years earlier … the tech revolution is happenin
g,” he said. Beyond opportunities to improve outcomes for individual patients, researchers are quickly seizing AI
to help solve population-level health challenges. But for AI to effectively tackle large-scale problems, academia
and industry will have to dissolve the boundaries between different scientific disciplines, said Dimitris Bertsima
s of Massachusetts Institute of Technology, one of the symposium’s keynote speakers. “AI-powered tools are now hel
ping dentists identify dental decay in patients up to five years earlier … the tech revolution is happening.” “Rea
l-world problems do not have [clear-cut] labels — global warming is not just physics, or engineering, or mathemati
cs. Medicine is not just biology, chemistry, or computer science,” said Bertsimas, the Boeing Professor of Operati
ons Research and associate dean of business analytics at MIT. “Multimodal data will increasingly be used across sc
ience, engineering, and medicine, and [AI] will become the predominant methodology for predictions and decision-ma
king across all fields.” At Harvard, cross-disciplinary teams are leveraging machine learning to identify patients

whose social determinants of health put them more directly in the path of climate-change-related impacts and a bevy of other risks to oral health. “Are exposures to wildfires impacting oral health? If they become more frequent, who’s most vulnerable and how do we act on this information?” asked Francesca Dominici, director of the Harvard Data Science Initiative at the T.H. Chan School of Public Health. She and a team of researchers are using AI to analyze satellite data, atmospheric chemistry models, and other factors, revealing which communities are most affected by increasingly frequent wildfires, extreme heat waves, and destructive storms. Reduced air quality from fires and higher temperatures from warming climate can cause mouths to be drier, making people more prone to oral disease and tooth decay. Increased psychological stress from extreme weather events can increase risk for teeth grinding and temporomandibular joint (TMJ) disorders. What’s more, “natural disasters can disrupt access to dental facilities and care,” Dominici added. Biomedical researchers are also deploying AI to speed up and optimize experiments, therapeutic discovery, and preclinical validation. “[AI is] generating, acquiring, harmonizing, and refining data, and it can generate hypotheses, as well as simulate experiments and downstream outcomes,” said Marinka Zitnik, an assistant professor of biomedical informatics at Harvard Medical School. It will revolutionize the way therapies are matched individually to patients, she said, and help design entirely new drugs and therapeutics. A survey of 1,600 biomedical researchers revealed that 25 percent of them feel AI will be essential to their studies within the decade, Zitnik added. She specializes in building knowledge graph AI models, which help contextualize and capture relationships within diverse sets of biomedical data. Her team has developed a knowledge graph AI model called TxGNN that describes 17,000 diseases using all available clinical and biomedical data. Once trained, it will be able to predict how well any given therapeutic might effectively treat a patient’s unique disease, and even be able to recommend new uses for FDA-approved medications. Participants used VR headsets to view a virtual dental practice in the workshop “The Future is Now — Innovate with Advanced AI.” Photo by Steve Gilbert On the industry side of dental care, two primary types of AI-assistive technologies are already making waves in dentists’ offices: platforms for patients, providers, and payers that focus on using AI to interpret and analyze imaging, and AI software that automates patient engagement, scheduling, and other time-consuming “back office” tasks for dental practices. “AI [products] on the market today are not decision-makers, they are helpers,” said Philippe Salah, co-founder and CEO of DentalMonitoring. Dental AI tools include products that allow dentists to remotely analyze patient-submitted oral photos submitted via smartphone. Imaging tools use AI to guide patients as they capture images of their teeth, and then can detect signs of declining oral health to flag for the dental care team. AI-guided 3D simulations of patients’ mouths help orthodontists accelerate the fittings and transition between bracers, aligners, and retainers. Some tools even enable patients to see AI-powered, life-like simulations of how their teeth, mouths, and faces will look after dental work or braces. “One of the biggest frustrations in my practice was knowing what my patients needed, but not having the tools or ability to communicate that in a way [that inspired my patients to] prioritize their dental care,” said Edward Zuckerberg, who owns a private practice and is the chief dental officer at both Keystone Bio and Viome. Although the upsides of AI are undoubtedly exciting, experts at the intersection of AI and dentistry agree that with progress must also come prudence. “I’m excited about the abundance of AI solutions [we’re talking about] here,” said Mariya Filipova, chief technology officer at CareQuest Innovation Partners. But, she cautioned, AI is not good at empathy, creativity, or imagination — all uniquely human factors that are critical to solving problems ethically. New technologies are often put into play before guidelines and policy have caught up. While more and more AI dental platforms are being cleared by the FDA for commercial access, that process “doesn’t regulate fairness and bias,” said Hawazin Elani, assistant professor of oral health policy and epidemiology at the Dental School. “We have to own where our data [that’s training AI systems] is coming from.” Fernanda Viégas, a keynote speaker at the symposium, warned that healthcare providers should not have to put blind faith in AI decision-making. Viégas, a principal scientist at Google and the Gordon McKay Professor of Computer Science at Harvard’s John A. Paulson School of Engineering and Applied Sciences, and her collaborators at Google have found that clinicians are more likely to adopt AI tools that don’t spit out automated results without sharing details about its analytic framework or the baseline data on which the system was trained. “In high-stakes situations” like making the correct diagnosis for a patient, Viégas said, “[we found] being able to engage with [AI] systems at meaningful levels mattered a lot” to healthcare providers. “It touches upon the need for trust, confidence, transparency… and these are all really hard things to accomplish [in an AI system]. As we start to deploy new [technologies], we will find these gaps [in user experience] that we need to design for.”

--- ARTICLE 9 ---
Title   : Cutting through the fog of long COVID
Date   : 2024-11-08
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/2024/11/cutting-through-the-fog-of-long-covid/

Content :
MGB Communications While earlier diagnostic studies have suggested that 7 percent of the population suffers from long COVID, a new AI tool developed by Mass General Brigham revealed a much higher 22.8 percent, according to the study. The AI-based tool can sift through electronic health records to help clinicians identify cases of long COVID. The often-mysterious condition can encompass a litany of enduring symptoms, including fatigue, chronic cough, and brain fog after infection from SARS-CoV-2. The algorithm used was developed by drawing de-identified patient data from the clinical records of nearly 300,000 patients across 14 hospitals and 20 community health centers in the Mass General Brigham system. The results, published in the journal Med, could identify more people who should be receiving care for this potentially debilitating condition. “Our AI tool could turn a foggy diagnostic process into something sharp and focused, giving clinicians the power to make sense of a challenging condition,” said senior author Hossein Estiri, head of AI Research at the Center for AI and Biomedical Informatics of the Learning Healthcare System (CAIBILS) at MGB and an associate professor of medicine at Harvard Medical School. “With this work, we may finally be able to see long COVID for what it truly is — and more importantly, how to treat it.” “With this work, we may finally be able to see long COVID for what it truly is — and more importantly, how to treat it.” For the purposes of their study, Estiri and colleagues defined long COVID as a diagnosis of exclusion that is also infection-associated. That means the diagnosis could not be explained in the patient’s unique medical record but was

associated with a COVID infection. In addition, the diagnosis needed to have persisted for two months or longer in a 12-month follow up window. The novel method developed by Estiri and colleagues, called “precision phenotyping,” sifts through individual records to identify symptoms and conditions linked to COVID-19 to track symptoms over time in order to differentiate them from other illnesses. For example, the algorithm can detect if shortness of breath results from pre-existing conditions like heart failure or asthma rather than long COVID. Only when every other possibility was exhausted would the tool flag the patient as having long COVID. “Physicians are often faced with having to wade through a tangled web of symptoms and medical histories, unsure of which threads to pull, while balancing busy caseloads. Having a tool powered by AI that can methodically do it for them could be a game-changer,” said Alaleh Azhir, co-lead author and an internal medicine resident at Brigham and Women’s Hospital, a founding member of the Mass General Brigham healthcare system. The new tool’s patient-centered diagnoses may also help alleviate biases built into current diagnostics for long COVID, said researchers, who noted diagnoses with the official ICD-10 diagnostic code for long COVID trend toward those with easier access to healthcare. The researchers said their tool is about 3 percent more accurate than the data ICD-10 codes capture, while being less biased. Specifically, their study demonstrated that the individuals they identified as having long COVID mirror the broader demographic makeup of Massachusetts, unlike long COVID algorithms that rely on a single diagnostic code or individual clinical encounters, skewing results toward certain populations such as those with more access to care. “This broader scope ensures that marginalized communities, often sidelined in clinical studies, are no longer invisible,” said Estiri. Limitations of the study and AI tool include that health record data the algorithm uses to account for long COVID symptoms may be less complete than the data physicians capture in post-visit clinical notes. Another limitation was the algorithm did not capture possible worsening of a prior condition that may have been a long COVID symptom. For example, if a patient had COPD that worsened before they developed COVID-19, the algorithm might have removed the episodes even if they were long COVID indicators. Declines in COVID-19 testing in recent years also makes it difficult to identify when a patient may have first gotten COVID-19. The study was limited to patients in Massachusetts. Future studies may explore the algorithm in cohorts of patients with specific conditions, like COPD or diabetes. The researchers also plan to release this algorithm publicly on open access so physicians and healthcare systems globally can use it in their patient populations. In addition to opening the door to better clinical care, this work may lay the foundation for future research into the genetic and biochemical factors behind long COVID’s various subtypes. “Questions about the true burden of long COVID — questions that have thus far remained elusive — now seem more within reach,” said Estiri. Support was given by the National Institutes of Health, National Institute of Allergy and Infectious Diseases (NIAID) R01AI165535, National Heart, Lung, and Blood Institute (NHLBI) OT2HL161847, and National Center for Advancing Translational Sciences (NCATS) UL1 TR003167, UL1 TR001881, and U24TR004111. J. Hügel’s work was partially funded by a fellowship within the IFI program of the German Academic Exchange Service (DAAD) and by the Federal Ministry of Education and Research (BMBF) as well by the German Research Foundation (426671079).

--- ARTICLE 10 ---
Title   : New journal, podcast take a closer look at artificial intelligence in medicine
Date   : 2023-04-10
Source : Harvard Gazette
URL   : https://news.harvard.edu/gazette/story/newsplus/new-journal-podcast-take-a-closer-look-at-artificial-intelligence-in-medicine/

Content :
iStock/ipopba Artificial intelligence (AI) has the power to change medicine and public health for the better, according to Harvard T.H. Chan School of Public Health’s Andrew Beam. But there’s also a need to proceed with caution in using this powerful tool, he said. Beam, assistant professor of epidemiology, is playing a key role in two companion efforts from NEJM Group—publisher of the New England Journal of Medicine—to explore the use of AI in medicine. He is serving, along with Harvard Medical School’s Arjun (Raj) Manrai, as co-deputy editor of NEJM AI, a journal that will publish its first articles this fall. Isaac (Zak) Kohane, founding chair of the Department of Biomedical Informatics in the Blavatnik Institute at Harvard Medical School, will be the editor-in-chief of NEJM AI. In addition, Beam and Manrai are co-hosting a podcast called NEJM AI Grand Rounds, which features guest experts to explore the deep issues at the intersection of AI, machine learning, and medicine. The launch of the journal and podcast come at a critical time, said Beam. “AI used to be hypothetical, in the future. But over the last two years, it has changed,” he said. “AI is here. Everyone knows about ChatGPT. But healthcare is different because patient lives are on the line. The stakes are higher in medical AI than in other kinds of AI. That really spurred me, Zak, and Raj to join this journal — to think seriously about a place to really kick the tires and understand when medical AI can be safely used.” The first three episodes of the podcast focus on three distinct areas of medicine. The first features Stanford Medicine’s Euan Ashley speaking about AI, genomics, and cardiology. In the second, Harvard Medical School’s Pranav Rajpurkar discusses AI and radiology. The third features Verily’s Lily Peng talking about AI in ophthalmology. Peter Lee, who leads Microsoft Research, speculates on how technologies like ChatGPT will transform medicine in the most recent episode. “[Ashley and his colleagues are] asking questions such as, ‘Can we get AI to read an echocardiogram in the same way that human cardiologists do?’” Beam said. “They’ve also found that AI can extract signals [from echocardiograms] that people can’t, such as biomarkers indicating anemia.” Peng spoke about the technology’s potential public health benefits. She noted that her team ran a trial in Thailand — where there aren’t a lot of trained ophthalmologists — that found that AI performed well in screenings for diabetic retinopathy in that population, and could potentially help people get diagnosed and treated more quickly. Aside from its promise, AI also poses serious challenges when used in medicine, Beam acknowledged. “There are structural biases baked into systems like ChatGPT because of the data they’re trained on — which is literally the entire Internet, which doesn’t always reflect our best selves,” said Beam. “So we have to make sure that AI doesn’t operationalize those biases on a large scale and hurt populations that are already at the fringes of the healthcare system.” — Karen Feldscher

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Tokenize Sentences and Remove Stopwords

Finally, to begin exploring trends within the dataset, basic text processing was applied to the content field (e.g., tokenization, stopword removal, lemmatization), followed by word frequency analysis. The top 20 most common words were visualized using a bar chart, offering insight into the dominant themes across the articles. Furthermore, a word cloud was generated to represent these frequently occurring terms in a visually engaging way, reinforcing the presence of key concepts such as "AI", "medicine", "patients", "technology", "treatment".

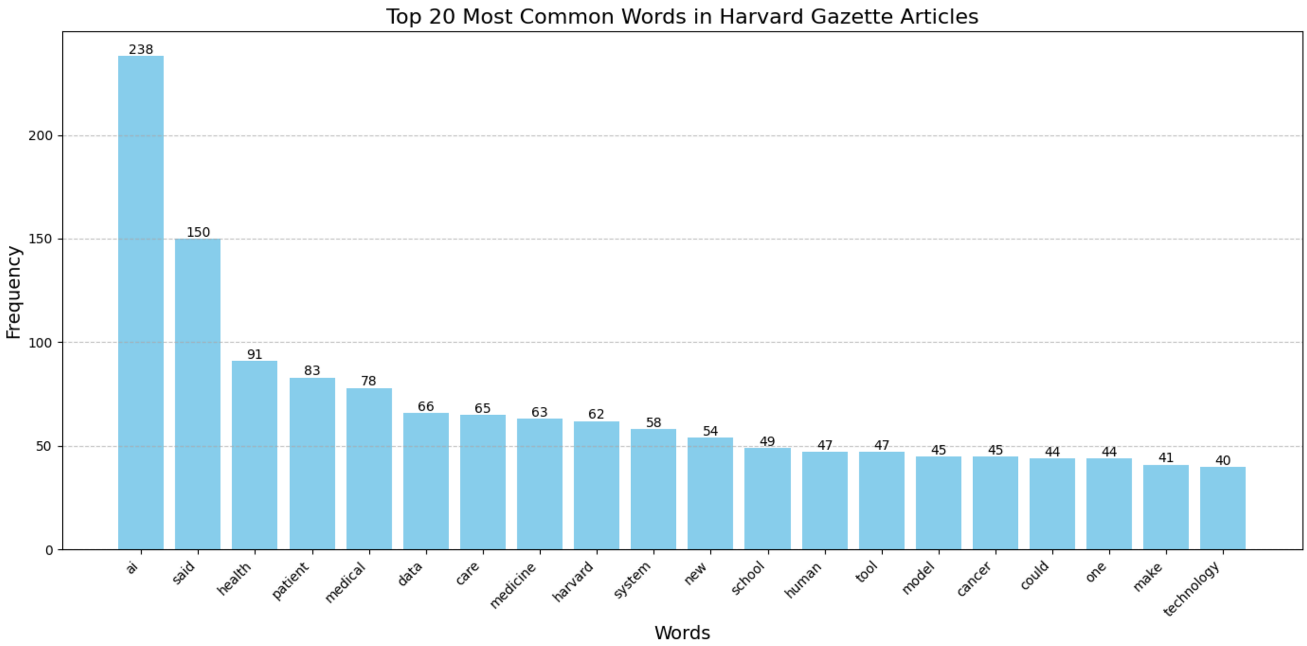
In [8]:
```
# Download NLTK data
nltk.download('punkt', quiet=True)
nltk.download('stopwords', quiet=True)
nltk.download('wordnet', quiet=True)
nltk.download('omw-1.4', quiet=True)

# Initialize stopwords and lemmatizer
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()

# Preprocessing function
def preprocess_text(text):
    # Lowercase and tokenize
    words = word_tokenize(text.lower())
    # Remove stopwords and non-alphanumeric, then lemmatize
    words = [lemmatizer.lemmatize(word) for word in words if word.isalnum() and word not in stop_words]
    return words

# Apply to the scraped 'content' column
df['processed_text'] = df['content'].apply(preprocess_text)

# Preview the result
print(df[['title', 'processed_text']])
```

```
                                                title  \
0                                     Machine healing
1                        The algorithm will see you now
2                     Great promise but potential for peril
3  New AI tool can diagnose cancer, guide treatme...
4                               AI revolution in medicine
5  Using AI to repurpose existing drugs for treat...
6  It’s inoperable cancer. Should AI make call ab...
7                  AI may be just what the dentist ordered
8                     Cutting through the fog of long COVID
9  New journal, podcast take a closer look at art...

                                       processed_text
0  [photo, illustration, judy, staff, alvin, powe...
1  [ashley, nunes, fellow, harvard, law, school, ...
2  [illustration, ben, boothman, christina, pazza...
3  [video, image, plus, ekaterina, pesheva, hm, c...
4  [illustration, ben, boothman, alvin, powell, h...
5  [identifies, possible, therapy, thousand, dise...
6  [rebecca, weintraub, brendel, director, harvar...
7  [kat, mcalpine, harvard, correspondent, dental...
8  [mgb, communication, earlier, diagnostic, stud...
9  [artificial, intelligence, ai, power, change, ...
```

Visualize Word Frequency Distribution

In [9]:
```
# Flatten list (one list with all words)
all_words = [word for tokens in df['processed_text'] for word in tokens]

# Create dictionary where keys are words and values are their counts
word_freq = Counter(all_words)

# Get most common words
top_n = 20
most_common_words = word_freq.most_common(top_n)

# Separate words and their counts into two tuples (words, counts)
words, counts = zip(*most_common_words)

# Plot bar chart with frequency
plt.figure(figsize=(14, 7))
```

```python
bars = plt.bar(words, counts, color='skyblue')
plt.title(f'Top {top_n} Most Common Words in Harvard Gazette Articles', fontsize=16)
plt.xticks(rotation=45, ha='right')
plt.xlabel('Words', fontsize=14)
plt.ylabel('Frequency', fontsize=14)
plt.grid(axis='y', linestyle='--', alpha=0.7)

# Add count labels above bars
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width() / 2, height + 1, str(height), ha='center', fontsize=10)

plt.tight_layout()
plt.show()
```



In [10]:

```python
# Generate word cloud
wordcloud = WordCloud(width=800, height=400, background_color='white').generate(' '.join(all_words))

# Plot word cloud
plt.figure(figsize=(14, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.show()
```

2. Text Classification

Once the articles dataset was successfully scraped and organized, the next step involved classifying each article into thematic categories to better structure the content and support downstream tasks like targeted summarization and retrieval. Instead of manually labeling each article, the project leveraged a zero-shot classification approach using a transformer-based language model.

Model and Rationale

The classification was performed using the `HuggingFace` pipeline API with the `valhalla/distilbart-mnli-12-3` model, which is a distilled version of BART fine-tuned on the Multi-Genre Natural Language Inference (MNLI) dataset. This makes it particularly effective for zero-shot learning tasks, where the model can assign texts to user-defined labels without any additional training.

This approach was selected for two key reasons:

1. Flexibility – It allows the use of custom, domain-specific labels tailored to this project.
2. Efficiency – It avoids the need to curate and annotate a training dataset, enabling rapid experimentation.

Thematic Labels

Three custom thematic categories were defined to reflect the most relevant areas of discourse around AI in healthcare:

- Clinical Applications & Tools: Articles focusing on diagnostic technologies, treatment systems, or AI-enabled tools in practice.
- Ethical & Patient Safety: Content discussing privacy concerns, decision-making risks, or moral considerations.
- Policy & Economic Impact: Pieces covering regulatory frameworks, cost implications, or broader social/political effects.

These labels were passed as candidate labels to the classifier, which evaluated the semantic fit of each article's content against the labels.

Classification Workflow

The entire body of each article was processed as input to the model. For each input, the classifier returned a ranked list of labels along with confidence scores. Only the top-ranked label (i.e., the one with the highest score) was retained for each article. The associated confidence score was also recorded to allow for transparency and potential filtering.

Although the model successfully categorized all articles, the confidence scores were generally modest (between 33%–39%), suggesting that thematic boundaries in healthcare discourse are nuanced and sometimes overlapping. This also reflects the limitation of using a generic NLI model on a specialized medical dataset. Nonetheless, the zero-shot approach offered a fast and effective solution for thematic tagging without the need for labeled training data.

In [11]:

```python
# Initialize zero-shot classification pipeline
classifier = pipeline("zero-shot-classification", truncation=True, model="valhalla/distilbart-mnli-12-3")

# Define candidate labels (thematic categories)
candidate_labels = [
    "Clinical Applications & Tools",
    "Ethical & Patient Safety",
    "Policy & Economic Impact"
    ]

themes = []
confidences = []

for content in df['content']:
    result = classifier(content, candidate_labels)
    # Get the top predicted label and score
    top_label = result['labels'][0]
    top_score = f"{result['scores'][0]:.2%}"

    themes.append(top_label)
    confidences.append(top_score)

df['theme'] = themes
df['confidence'] = confidences
```

```python
df[['title', 'theme', 'confidence']]
```

```
Device set to use cpu
```

Out[11]:

	title	theme	confidence
0	Machine healing	Ethical & Patient Safety	35.27%
1	The algorithm will see you now	Policy & Economic Impact	34.18%
2	Great promise but potential for peril	Ethical & Patient Safety	33.61%
3	New AI tool can diagnose cancer, guide treatme...	Clinical Applications & Tools	33.91%
4	AI revolution in medicine	Policy & Economic Impact	36.94%
5	Using AI to repurpose existing drugs for treat...	Policy & Economic Impact	33.59%
6	It's inoperable cancer. Should AI make call ab...	Policy & Economic Impact	33.92%
7	AI may be just what the dentist ordered	Clinical Applications & Tools	34.32%
8	Cutting through the fog of long COVID	Policy & Economic Impact	34.25%
9	New journal, podcast take a closer look at art...	Ethical & Patient Safety	39.57%

3. Information Retrieval & RAG System

Once the articles had been categorized, the next objective was to build a Retrieval-Augmented Generation (RAG) system that could answer natural language questions by grounding its responses in the scraped article content. This system combines vector-based document retrieval with large language models (LLMs) for context-aware question answering, allowing the user to interact with the corpus as if it were a live knowledge base.

Vector Embedding & Semantic Search

The first step involved transforming each article into a vector representation, capturing its semantic content. This was achieved using the `all-MiniLM-L6-v2` model from `SentenceTransformer`, which encodes each article into a 384-dimensional embedding. This model strikes a balance between speed and semantic accuracy, making it ideal for projects that require near real-time vector search with limited hardware.

These embeddings were stored in a FAISS index, a high-performance similarity search library optimized for dense vector search. FAISS was used in flat L2 mode (`IndexFlatL2`), where all vectors are stored and searched exhaustively, a feasible choice given the small size of the dataset. This enabled fast retrieval of the top k most relevant articles for any user query, based on semantic similarity.

Prompt Engineering & Response Generation

To generate answers, a prompt template was created that included:

- A clear system role ("You are a helpful assistant specialized in medical and academic topics")
- A structured context block containing shortened text chunks from top-matching articles
- A reminder to cite sources using numeric references
- The user's question

This setup allows the language model to generate answers that are both context-grounded and explainable, with explicit references to supporting texts.

Two retrieval functions were created to support answer generation:

- One using `GPT-2` via `Hugging Face`'s pipeline("text-generation"), with prompts formatted manually to include context chunks, retrieval references, and assistant instructions.

- Another using `GPT-4o` via the OpenAI API, structured around a dedicated system_message that outlines constraints such as avoiding hallucination, using only retrieved sources, and maintaining a professional tone.

In both methods, the top 3 most relevant articles are retrieved based on vector similarity. Their content is truncated for prompt compactness and injected into the LLM prompt alongside the user's question.

In [12]:

```python
# 1. Load articles
# df = pd.read_csv("ai_healthcare_articles.csv")

# 2. Load embedding model
embedding_model = SentenceTransformer('all-MiniLM-L6-v2')

# 3. Compute embeddings for all articles
print("Encoding article contents...")
texts = df['content'].fillna("").tolist()
titles = df['title'].fillna("").tolist()
embeddings = embedding_model.encode(texts, convert_to_numpy=True)

# 4. Create or load FAISS index
dimension = embeddings.shape[1]
index = faiss.IndexFlatL2(dimension)
index.add(embeddings)
print(f"✅ FAISS index created and populated with {index.ntotal} vectors.")

# 5. Save index
faiss.write_index(index, "embeddings.index")
print("✅ FAISS index saved to 'embeddings.index'")

# 6. Define QA function with prompt engineering
def retrieve_with_prompt(query, model, index, texts, titles, k=3, max_context_chars=300):
    query_embedding = model.encode([query], convert_to_numpy=True)
    distances, indices = index.search(query_embedding, k=k)

    top_chunks = [texts[i] for i in indices[0]]
    top_titles = [titles[i] for i in indices[0]]

    # Create context with shortened text for readability
    context = "\n\n".join(
        f"[{i+1}] {textwrap.shorten(text.replace('\n', ' '), width=max_context_chars)}"
        for i, text in enumerate(top_chunks)
    )

    # Prompt engineering
    prompt = (
        "You are a helpful assistant specialized in medical and academic topics.\n"
        "Based on the context sections below, provide a clear and concise answer to the user’s question.\n"
        "Reference the relevant context sections using their numbers (e.g., [1], [2]).\n\n"
        f"Context:\n{context}\n\n"
        f"Question: {query}\nAnswer:"
    )

    # Generate answer using GPT-2
    generator = pipeline("text-generation", model="gpt2", tokenizer="gpt2")
    response = generator(prompt, max_new_tokens=200, do_sample=True, top_k=50, temperature=0.8)[0]['generated_tex
    answer = response.replace(prompt, "").strip()

    return answer, top_titles

# 7. Example QA
example_query = "How is AI used in diagnosing diseases?"
answer, sources = retrieve_with_prompt(example_query, embedding_model, index, texts, titles)

# 8. Print output with formatting
print(f"\n🧠 Query:\n{textwrap.fill(example_query, width=100)}")
print(f"\n💬 Answer:\n{textwrap.fill(answer, width=100)}")
print("\n📚 Source Articles:")
for i, src in enumerate(set(sources), 1):
    print(f"{i}. {src}")
```

```
Encoding article contents...
✅ FAISS index created and populated with 10 vectors.
✅ FAISS index saved to 'embeddings.index'
```

```
Device set to use cpu
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
```

```
🧠 Query:
How is AI used in diagnosing diseases?

💬 Answer:
[1] MGB Communications While earlier diagnostic studies have suggested that 7 percent of the
population suffers from long COVID, a new AI tool developed by Mass General Brigham revealed a much
higher 22.8 percent, according to the study. The AI-based tool can sift through electronic health
records [...]  [2] Illustration by Ben Boothman Alvin Powell Harvard Staff Writer Third in a series
that taps the expertise of the Harvard community to examine the promise and potential pitfalls of
the coming age of artificial intelligence and machine learning. The news is bad: “I’m sorry, but you
have cancer.” [...]  [3] Identifies possible therapies for thousands of diseases, including ones
with no current treatments Ekaterina Pesheva HMS Communications Much of the history of the human
body has been thought of as a complex system that functions mostly in concert with other parts of
our bodies. However, research and development of new therapies in these bodies is not well known

📚 Source Articles:
1. Using AI to repurpose existing drugs for treatment of rare diseases
2. AI revolution in medicine
3. Cutting through the fog of long COVID
```

Storing and Loading the OpenAI API Key as an Environment Variable

For secure and maintainable handling of credentials, the OpenAI API key was stored in a `.env` file and loaded into the environment. First, a dedicated Python virtual environment was created for the project to isolate dependencies and maintain a clean development environment. The `.env` file was created using the terminal command `echo OPENAI_API_KEY=your_api_key > .env`, which stored the API key in the appropriate format. To verify that the file was created correctly, `cat .env` (or type `.env` on Windows) was executed, which returned the expected output: `OPENAI_API_KEY=your_api_key`. Additionally, its existence was confirmed in Python using `print(os.path.isfile('.env'))`, which returned `True`.

In [13]:
```python
# Check if the .env file exists
print(os.path.isfile('.env'))
```

```
True
```

Within the pipeline, the key was accessed securely using `os.getenv("OPENAI_API_KEY")`, ensuring it was never hard-coded into the script. This key was then passed to the OpenAI client for authenticated access to the GPT-4o model. By using this approach within a virtual environment, the project remains both secure and portable, enabling seamless collaboration and deployment without exposing sensitive credentials.

In [14]:
```python
# Initialize OpenAI client
client = OpenAI(api_key = os.getenv("OPENAI_API_KEY"))

# Load embedding model
embedding_model = SentenceTransformer('all-MiniLM-L6-v2')

# Compute embeddings for all articles
texts = df['content'].fillna("").tolist()
titles = df['title'].fillna("").tolist()

print("Encoding article contents...")
embeddings = embedding_model.encode(texts, convert_to_numpy=True)

# Create FAISS index
dimension = embeddings.shape[1]
index = faiss.IndexFlatL2(dimension)
index.add(embeddings)
print(f"✅ FAISS index created with {index.ntotal} vectors.")

# Save the index if needed
faiss.write_index(index, "embeddings.index")
print("✅ FAISS index saved to 'embeddings.index'")

# Define system message for the assistant
system_message = """
You are an intelligent assistant designed to answer user questions
accurately and concisely by using the provided content.
Your goal is to prioritize information from the documents and resources available to you,
ensuring that all responses are grounded in the retrieved content.

Key Guidelines:
1. Use only the provided content to answer questions. Do not make assumptions or include information that is not
```

```
2. If the content does not provide sufficient information to answer the question, politely inform the user that t
3. Present responses in a clear and professional manner, tailored to the user's question.
4. When appropriate, provide additional context or clarification from the retrieved content, but avoid unnecessar
5. Use natural and conversational language to ensure the response is user-friendly.

Your purpose is to assist users in understanding and utilizing the provided information effectively.
"""

def retrieve_and_answer(user_question, model, index, texts, titles, k=3, max_context_chars=500):
    # 1. Embed the user query
    query_embedding = embedding_model.encode([user_question], convert_to_numpy=True)

    # 2. Search for the top k relevant documents
    distances, indices = index.search(query_embedding, k=k)

    # 3. Get the matching article texts and titles
    matched_texts = [texts[i] for i in indices[0]]
    matched_titles = [titles[i] for i in indices[0]]

    # 4. Build the prompt content with clear references and truncated text
    context = "\n\n".join(
        f"[{i+1}] {textwrap.shorten(text.replace('\n', ' '), width=max_context_chars)}"
        for i, text in enumerate(matched_texts)
    )

    # 5. Construct the user prompt for the LLM
    prompt = f"""
The following content is provided to help answer user questions:

Content:
{context}

User Question:
{user_question}

Please use this content to answer the user's question as accurately as possible.
Remember to only use the information explicitly mentioned in the content and avoid adding any
outside knowledge or assumptions. Also please use clever formatting for readability.
"""

    # 6. Call OpenAI API
    completion = client.chat.completions.create(
        model=model,
        messages=[
            {"role": "system", "content": system_message},
            {"role": "user", "content": prompt},
        ],
        temperature=0.7,
        max_tokens=500,
    )

    answer = completion.choices[0].message.content.strip()

    return answer, matched_titles


# Example usage
user_question = "How is AI used in diagnosing diseases?"
answer, sources = retrieve_and_answer(user_question, model="gpt-4o", index=index, texts=texts, titles=titles)

print("\n🧠 Query:\n", user_question)
print("\n💬 Answer:\n", answer)
print("\n📚 Source Articles:")
for i, title in enumerate(sources, 1):
    print(f"{i}. {title}")
```

```
Encoding article contents...
✅ FAISS index created with 10 vectors.
✅ FAISS index saved to 'embeddings.index'

🧠 Query:
 How is AI used in diagnosing diseases?

💬 Answer:
 AI is being utilized in diagnosing diseases in the following ways:

1. **Long COVID Diagnosis**:
   - An AI tool developed by Mass General Brigham is used to identify cases of long COVID by analyzing electronic
health records. This tool revealed that 22.8% of the population might suffer from long COVID, which is a higher es
timate than previous studies suggesting 7% ([1]).

2. **Cancer Diagnosis**:
   - Advances in AI allow doctors to compare a patient's cancer case with thousands of other cases. This compariso
n can help in understanding the disease better and potentially improve diagnosis and treatment decisions ([2]).

These examples illustrate how AI is leveraged to enhance the accuracy and efficiency of diagnosing various health
conditions.

📚 Source Articles:
1. Cutting through the fog of long COVID
2. AI revolution in medicine
3. Using AI to repurpose existing drugs for treatment of rare diseases
```

Two-Model Comparison: GPT-2 vs GPT-4o

Model 1: GPT-2 (Local Hugging Face Model)

Initially, the system was tested using the open-source `gpt2` model via `HuggingFace`'s text generation pipeline. Despite being accessible and fast, GPT-2 has significant limitations:

- It has no built-in chat memory or instruction-following capacity
- It produces verbose, occasionally hallucinatory responses
- It lacks the fine-tuned control required for structured or sourced answers

In fact, the answer generated by GPT-2 to the query “How is AI used in diagnosing diseases?” seemed to be in parts overly generalized, verbose and occasionally inconsistent. This illustrates GPT-2’s limitations in handling fact-based, high-stakes content, especially when precision and traceability are crucial.

Model 2: GPT-4o (via OpenAI API)

The same user query was passed to this model using a refined prompt that emphasized:

- Strict use of only the provided content
- Clear formatting (e.g., bullet points, headings)
- Professional tone and conversational language

The responses generated by GPT-4o were notably more accurate, structured, and context-aware than those from GPT-2. For example, when asked “How is AI used in diagnosing diseases?”, GPT-4o returned a segmented answer citing long COVID detection, cancer diagnosis, and AI imaging tools—each tied explicitly to a source article.

This comparison underscores the advantage of modern instruction-tuned models and also highlights the value of thoughtful prompt design in maintaining LLM reliability.

4. Summarization & Prompt Evaluation

The final stage involved summarizing each article using three distinct prompting strategies. A custom `build_prompt_strategy()` function defined different summarization modes:

- Role-based prompt: instructs the model to behave like a scientific editor writing for a research bulletin.
- Format constraint prompt: restricts the output to exactly three bullet points under 20 words each.
- Fallback prompt: asks the model to simplify technical content or highlight only the key takeaways when the article is unclear.

Each strategy was passed to GPT-4o using the OpenAI API in a structured chat-completion format. This design allowed consistent control over tone, output length, and structure. The model was able to successfully adapt its behavior to each prompt. For instance, role-based summaries included academic terminology and longer paragraphs, while bullet-pointed summaries emphasized clarity and brevity. Fallback summaries offered a balance of simplification and informativeness, crucial for general audiences.

To evaluate summarization quality, the pipeline implemented a second LLM-based system that assessed each output across four criteria:

- Coverage (are all important ideas present?)
- Conciseness (is the summary brief and to the point?)
- Coherence (is it logically structured?)
- Faithfulness (is it accurate to the original content?)

Each summary was assigned scores in structured JSON format, enabling objective comparison between strategies. Results showed that role-based prompts achieved the highest scores across all four criteria, while fallback prompts provided a strong balance of brevity and accuracy, and format-constrained bullet summaries, though clear, lagged in coverage and faithfulness.

In [15]:
```
# Load the OpenAI API key from .env
load_dotenv()
client = OpenAI(api_key = os.getenv("OPENAI_API_KEY"))

# Define different prompt strategies
def build_prompt_strategy(content, strategy="default"):
    content = content.strip()

    if len(content) < 50:
        return "The content is too short to generate a meaningful summary."

    if strategy == "role":
        return f"""
You are a scientific editor. Please summarize the following article for a research bulletin:

\"\"\"{content}\"\"\"

Summary:
"""

    elif strategy == "format_constraint":
        return f"""
Summarize the following article in exactly 3 bullet points. Each bullet should be under 20 words:

\"\"\"{content}\"\"\"

Summary:
-
-
-
"""

    elif strategy == "fallback":
        return f"""
If the content is too technical, simplify it. If it's unclear, summarize only the key messages:

\"\"\"{content}\"\"\"

Summary:
"""

    else:  # default
        return f"""
Summarize the following article in 2-3 concise sentences:

\"\"\"{content}\"\"\"

Summary:
"""

# Generate summaries
def generate_summaries(df, strategy="default", model="gpt-4o"):
    summaries = []
```

```python
    for content in df['content']:
        prompt = build_prompt_strategy(content, strategy)

        if "too short" in prompt:
            summaries.append("⚠️ Content too short to summarize.")
            continue

        response = client.chat.completions.create(
            model=model,
            messages=[
                {"role": "system", "content": "You are a helpful assistant that summarizes academic articles."},
                {"role": "user", "content": prompt}
            ],
            temperature=0.5,
            max_tokens=250
        )

        summary = response.choices[0].message.content.strip()
        summaries.append(summary)

    return summaries

# Generate summaries for each strategy
df["summary_role"] = generate_summaries(df, strategy="role")
df["summary_bullets"] = generate_summaries(df, strategy="format_constraint")
df["summary_fallback"] = generate_summaries(df, strategy="fallback")

# Optional: Save to file
# df.to_csv("summarized_articles.csv", index=False)

# Display the results
for i, row in df.iterrows():
    print(f"\n📝 Article {i+1}:")
    print(f"\n🔹 Role Summary:\n{row['summary_role']}")
    print(f"\n🔹 Bullet Summary:\n{row['summary_bullets']}")
    print(f"\n🔹 Fallback Summary:\n{row['summary_fallback']}")
```

📝 Article 1:

🔹 Role Summary:
The article discusses the transformative impact of artificial intelligence (AI), particularly large language models (LLMs) like ChatGPT, on the medical field. Adam Rodman, a Harvard Medical School assistant professor, highlights how AI tools like OpenEvidence, which quickly summarize medical literature, have drastically improved efficiency in patient care. Experts predict that AI will reshape various aspects of healthcare, from doctor-patient interactions to medical education, potentially increasing efficiency, reducing errors, and alleviating administrative burdens. However, concerns remain about biases in data sets, AI's tendency to generate false information, and the risk of insufficiently ambitious integration of AI into healthcare systems.

Isaac Kohane, chair of Harvard’s Department of Biomedical Informatics, emphasizes AI's potential to address inefficiencies in the U.S. healthcare system, highlighting a case where AI correctly diagnosed a condition missed by multiple doctors. Despite AI's capabilities, studies show that its integration with doctors' work has not significantly improved diagnostic accuracy, suggesting that training and usage strategies need refinement.

AI also offers opportunities to improve patient safety and reduce clinician burnout by automating tasks like note-taking. However, experts caution that AI should not replace human oversight, especially given the biases rooted in historical medical data. Initiatives like the MIM

🔹 Bullet Summary:
- Adam Rodman uses an AI app, OpenEvidence, to quickly access medical literature, enhancing patient care efficiency.
- Experts predict AI will reshape medicine, improving efficiency and decision-making but may reinforce existing biases.
- AI's role in medicine includes potential benefits in notetaking, diagnosis, and research, but human oversight remains crucial.

🔹 Fallback Summary:
The article discusses the transformative potential of artificial intelligence (AI), particularly large language models (LLMs) like ChatGPT, in the field of medicine. Adam Rodman, a doctor and assistant professor at Harvard Medical School, uses a smartphone app called OpenEvidence to access medical literature quickly, enhancing patient care. The adoption of AI in medicine is expected to improve efficiency, reduce errors, and ease administrative burdens, potentially reshaping doctor-patient interactions, healthcare administration, and medical education. However, there are concerns about AI's potential to reinforce existing biases and its tendency to "hallucinate" or generate false information.

Isaac Kohane, chair of Harvard's Department of Biomedical Informatics, highlights the capabilities of AI, noting its potential to address inefficiencies in the U.S. healthcare system. AI is not expected to replace doctors but to complement them, potentially transforming how medical care is delivered. Despite AI's promise, challenges remain, such as ensuring diverse data representation to avoid biases.

AI's role extends to notetaking and summarization, which could reduce doctors' paperwork and improve doctor-patient relationships. Nevertheless, AI systems are not yet reliable enough to operate without human oversight. The article also discusses AI's potential in medical research, such as

📝 Article 2:

🔹 Role Summary:
The Harvard Global Health Institute recently hosted a symposium titled "Hype vs. Reality: The Role of AI in Global Health," where experts from academia, healthcare, and industry convened to discuss the potential and challenges of using artificial intelligence (AI) in healthcare. The event highlighted successful AI applications, such as Google's diabetic retinopathy screening tool and startups like Aindra Systems and Ubenwa, which use AI for cancer and birth asphyxia detection, respectively. However, speakers cautioned against overly optimistic views of AI, emphasizing the need for robust medical infrastructure and appropriate data to support AI tools. Challenges such as data quality, bias, and ensuring equitable access to AI technologies were discussed. The symposium underscored the importance of integrating AI as part of a broader healthcare strategy rather than viewing it as a standalone solution. Participants noted that while AI has the potential to transform healthcare, especially in resource-poor settings, it should complement existing efforts to address healthcare disparities and should not divert resources from critical infrastructure investments.

🔹 Bullet Summary:
- AI in healthcare shows promise for diagnosing diseases like diabetic retinopathy and leprosy, but challenges remain.
- Experts warn AI should not divert funds from essential healthcare infrastructure and must be used to solve real problems.
- AI's potential benefits must be accessible to all, not just the wealthy, to avoid increasing inequality in healthcare.

🔹 Fallback Summary:
The article discusses a conference at Harvard where experts gathered to discuss the role of artificial intelligence (AI) in global health care. AI has shown promise in diagnosing conditions like diabetic retinopathy and leprosy, potentially transforming health care in areas with limited medical infrastructure. Despite its potential, there are concerns that overestimating AI's capabilities could divert funds from necessary health care investments. Experts emphasized that AI should address specific health care needs rather than being a solution in search of a proble

m. Challenges include ensuring the quality and availability of data for training AI systems and preventing AI from exacerbating inequalities by benefiting only the wealthy. The conference highlighted successful AI applications, like Google's diabetic retinopathy screening tool and Ubenwa's infant cry analysis for detecting birth asphyxia. However, practical issues like data entry errors and connectivity problems can hinder AI implementation in the field. Speakers argued for innovative approaches to health care delivery, especially in resource-poor settings, and stressed the importance of ensuring AI benefits everyone, not just the affluent.

📝 Article 3:

🔹 Role Summary:
The article, part of a series exploring artificial intelligence (AI) and machine learning, discusses the transformative impact of AI across various industries and the ethical concerns it raises. AI has become integral to sectors like healthcare, banking, retail, and manufacturing, promising increased efficiency and cost reduction. However, concerns about societal harm, bias, and lack of regulation persist. AI's ability to analyze data quickly aids in strategic decision-making and product development, notably in pharmaceuticals. In employment, AI supports tasks previously done by humans, potentially enhancing productivity. For small businesses, AI offers insights into financial operations and improves lending processes. Yet, ethical concerns about privacy, bias, and the role of human judgment remain significant. Experts argue for tighter regulation, but there's little consensus on how to achieve this. The U.S. lags in tech regulation, unlike the EU, which is more advanced in data privacy laws. The article underscores the need for ethical considerations and education on AI's societal impact, advocating for more informed oversight and responsible use of technology.

🔹 Bullet Summary:
- AI's rapid growth is transforming industries, but concerns about bias and lack of regulation persist.
- AI can enhance productivity and decision-making in businesses, especially small ones, by providing real-time insights.
- Ethical concerns about AI include privacy, discrimination, and the need for human judgment in decision-making.

🔹 Fallback Summary:
This article is the second in a series exploring the promise and challenges of artificial intelligence (AI) and machine learning. AI, initially a tool for high-level STEM research, is now integral to various industries, including healthcare, banking, retail, and manufacturing. Its potential to improve efficiency and reduce costs is significant, but concerns about societal harm, such as bias and lack of transparency, are growing.

AI's rapid growth is fueled by powerful computers and large datasets, with machine learning playing a key role. It is transforming industries by streamlining processes like sourcing materials, strategic decision-making, and product development. In healthcare, AI aids in data analysis and diagnosis, while in employment, it assists in resume screening and interview analysis, creating "hybrid" jobs that enhance human productivity.

Despite AI's benefits, concerns about bias and discrimination persist. Algorithms can perpetuate existing societal biases, especially if not carefully managed. In lending, AI could improve access to capital for small businesses but may also replicate past discriminatory practices unless carefully regulated.

There is debate over how to regulate AI. While some argue for tighter regulation, others believe existing industry-specific bodies could oversee AI's impact. The U.S. lags behind in tech regulation compared to the European Union. Experts like

📝 Article 4:

🔹 Role Summary:
Scientists at Harvard Medical School have developed a versatile AI model, CHIEF (Clinical Histopathology Imaging Evaluation Foundation), capable of performing a wide range of diagnostic tasks across multiple cancer types. Detailed in *Nature*, this AI system surpasses existing models by being trained on 19 cancer types and achieving nearly 94% accuracy in cancer detection. CHIEF excels in predicting patient outcomes and identifying genomic variations without the need for extensive DNA sequencing, offering a cost-effective alternative. It was tested on over 19,400 images from diverse global datasets and demonstrated superior performance in predicting tumor characteristics, patient survival, and treatment responses. The model's adaptability allows it to work effectively across different clinical settings and tissue acquisition methods. CHIEF also identified novel tumor features linked to patient survival, providing valuable insights into cancer prognosis. The researchers aim to further refine and expand CHIEF's capabilities, with the potential to enhance global cancer diagnostics significantly. The study received support from various institutions, including the National Institute of General Medical Sciences and Harvard Medical School.

🔹 Bullet Summary:
- Harvard scientists developed CHIEF, a versatile AI model for diagnosing and predicting outcomes across 19 cancer types.
- CHIEF achieved 94% accuracy in cancer detection and outperformed other AI methods in predicting patient outcomes.
- It accurately identified genomic variations and treatment responses, offering a cost-effective alternative to DNA sequencing.

🔹 Fallback Summary:
Scientists at Harvard Medical School have developed a new AI model, named CHIEF, designed to perform a wide range of diagnostic tasks for multiple cancer types, similar to how ChatGPT functions in language processing. Unlike current AI systems, which are often limited to specific tasks or cancer types, CHIEF is versatile and has been tested on 19 different cancers. It can detect cancer, predict patient outcomes, and identify treatment responses with hig

h accuracy.

CHIEF was trained on millions of images and can analyze both specific image sections and whole slides, allowing it to interpret cancer images more holistically. It achieved nearly 94% accuracy in cancer detection and significantly outperformed existing AI methods in various tasks, including predicting genomic variations and patient survival.

The model can identify specific tumor characteristics linked to patient survival and treatment response, offering a potential alternative to genomic sequencing, which is costly and time-consuming. CHIEF also demonstrated high accuracy in predicting mutations related to treatment responses across multiple cancer types.

Researchers believe that if further validated, CHIEF could help identify patients who might benefit from experimental treatments, especially in regions where genomic profiling is not readily available. The model's adaptability across different clinical settings and techniques enhances its potential for widespread use in cancer diagnostics.

📝 Article 5:

🔹 Role Summary:
The article explores the transformative potential and challenges of integrating artificial intelligence (AI) into the field of medicine. It highlights how AI can revolutionize personalized medicine by analyzing vast amounts of data to tailor treatments to individual patients, potentially improving outcomes and reducing costs. AI's capabilities in medical imaging, such as diagnosing diseases with high accuracy, are emphasized as areas where it is already showing promise.

However, the article also warns of the risks associated with AI, including the potential for misdiagnosis, perpetuating biases present in training data, and increasing healthcare costs if not properly implemented. Experts stress the importance of designing AI systems thoughtfully, incorporating insights from diverse fields such as ethics, sociology, and behavioral economics to address these challenges.

The article discusses the potential of AI to fill gaps in healthcare, especially in underserved regions, by providing diagnostic support and specialist insights remotely. It also touches on AI's role in managing the COVID-19 pandemic, noting both successes and shortcomings due to data access issues.

Further, the article mentions ongoing efforts to improve AI's interpretability and its integration into everyday medical practice, emphasizing the need for AI to complement rather than replace human decision-making. The importance of continuous evaluation and interdisciplinary collaboration is highlighted to ensure AI systems are effective and equitable.

🔹 Bullet Summary:
- AI in medicine promises personalized treatments and improved efficiency but requires careful implementation to avoid biases and errors.
- AI excels in medical imaging, potentially surpassing human experts, but its integration must consider ethical and societal impacts.
- AI applications in healthcare can improve global access to quality care, but challenges like data biases and transparency remain.

🔹 Fallback Summary:
The article explores the transformative potential and challenges of integrating artificial intelligence (AI) into medicine, highlighting insights from Harvard experts. Key points include:

1. **Personalized Medicine:** AI can analyze vast amounts of medical data to offer personalized treatment plans, potentially improving outcomes for diseases like cancer.

2. **Efficiency and Error Reduction:** Properly designed AI could make healthcare systems more efficient, reduce costs, and minimize medical errors.

3. **Challenges and Risks:** AI systems can misdiagnose if poorly designed. Bias in data can lead to flawed outcomes, and AI-driven systems may increase costs if not carefully implemented.

4. **AI in Medical Imaging:** AI shows promise in fields like radiology and pathology, where it can match or surpass human expertise in image recognition tasks.

5. **COVID-19 Response:** AI had limited impact during the pandemic due to data access issues, but it showed potential in early detection and vaccine development.

6. **Behavioral Change and Ethics:** AI's success depends on addressing human behavior and ethical concerns. Algorithms must be transparent and free from bias.

7. **Interdisciplinary Approach:** Collaboration across fields like ethics, sociology, and psychology is crucial to understand AI's real-world impact and ensure its responsible use.

8

📝 Article 6:

🔹 Role Summary:

A new artificial intelligence tool, TxGNN, developed by scientists at Harvard Medical School, offers a promising approach to identifying potential therapies for over 17,000 diseases, many of which currently lack treatments. This AI model is specifically designed to repurpose existing medicines, providing a faster and more cost-effective method than developing new drugs from scratch. The tool, which is the first of its kind to target rare and neglected diseases, identifies drug candidates from nearly 8,000 existing medicines and predicts possible side effects and contraindications, outperforming current AI models by nearly 50% in identifying drug candidates and 35% in predicting contraindications. By analyzing shared genomic features across multiple diseases, TxGNN can extrapolate treatment possibilities from well-understood conditions to those that are poorly understood. The model has been validated using 1.2 million patient records and has demonstrated the ability to reason similarly to a human clinician. The researchers emphasize that while the tool can expedite drug repurposing, further evaluation is needed for dosing and delivery. The AI model is available for free to encourage its use in discovering new therapies, particularly for rare and neglected diseases. The research was supported by various grants and collaborations with pharmaceutical companies and foundations.

🔹 Bullet Summary:
- TxGNN, an AI tool, identifies drug candidates for over 17,000 diseases, many lacking treatments, using existing medicines.
- The model is 50% better at identifying drugs and 35% more accurate in predicting contraindications than leading AI models.
- It uses shared genomic features across diseases, offering a strategic approach to drug repurposing beyond serendipity.

🔹 Fallback Summary:
Researchers at Harvard Medical School have developed an AI tool called TxGNN to identify potential therapies for over 17,000 diseases, including many rare and untreated conditions. This tool repurposes existing medicines, which is faster and more cost-effective than creating new drugs. TxGNN is notable for its ability to handle a vast number of diseases, making it the largest AI model of its kind. It predicts drug candidates and their side effects more accurately than existing models, improving drug repurposing by identifying shared features across diseases.

The AI tool was tested on extensive data, including patient records and genomic information, and demonstrated its ability to suggest treatments for diseases it had not been specifically trained on. It also provides explanations for its decisions, increasing transparency and physician confidence. While the identified therapies need further evaluation, TxGNN offers a promising approach to addressing the global burden of rare and neglected diseases. The researchers are collaborating with various foundations to explore treatment possibilities further.

📝 Article 7:

🔹 Role Summary:
The article discusses the implications of using large-language AI models in patient care, particularly in end-of-life decision-making. Rebecca Weintraub Brendel, director of Harvard Medical School’s Center for Bioethics, highlights the potential for AI to assist in medical decision-making by processing vast amounts of information objectively. However, she emphasizes the importance of maintaining human involvement in decisions of significant consequence to ensure respect for patient values and individuality. Brendel raises ethical questions about AI's role in health care, such as its impact on the physician-patient relationship, the potential for bias, and the responsibilities of healthcare professionals in a technologically advanced age. The discussion also touches on the need for moral leadership in medicine and the importance of aligning AI developments with values of equity, justice, and global health. Brendel advocates for careful consideration of AI's role in healthcare, ensuring that technological advancements do not overshadow the human elements of care, particularly in under-resourced settings.

🔹 Bullet Summary:
- Large-language AI models may expand patient care roles but raise ethical concerns regarding end-of-life decision-making.
- AI's diagnostic capabilities can aid in patient care, but human oversight remains crucial for ethical decision-making.
- Ensuring AI aligns with healthcare values like justice and respect is vital, especially in under-resourced settings.

🔹 Fallback Summary:
Rebecca Weintraub Brendel, director of Harvard Medical School’s Center for Bioethics, discusses the implications of using large-language AI models in patient care, particularly in end-of-life decision-making. AI is already aiding in analyzing medical imaging, but its potential expansion into broader patient care raises ethical questions. Key points include:

1. **End-of-Life Decisions**: Decision-making should align with patients' wishes, provided they are competent and medically appropriate. AI might assist in understanding patient conditions and potential outcomes but should not replace human judgment.

2. **AI's Role**: AI can process vast amounts of information objectively, which could aid in prognosis and treatment decisions. However, it should not dictate decisions based solely on statistical outcomes, as human context and values are crucial.

3. **Human Element**: Maintaining human involvement in significant medical decisions is essential. While AI can outperform humans in diagnostics, the interpretation and application of this information must consider human values and empathy.

4. **Ethical Considerations**: The integration of AI in healthcare necessitates discussions on moral leadership, equity, and justice. Ensuring AI aligns with societal values and addresses global health disparities is vital.

5. **Future of Medicine**: The role of healthcare professionals may

📝 Article 8:

🔹 Role Summary:
The inaugural Global Symposium on AI & Dentistry at Harvard highlighted the transformative potential of AI in oral healthcare. Experts discussed AI's ability to revolutionize dental practices, improve disease detection, and enhance access to care. The symposium featured over 65 research projects, including AI-driven device prototypes and patient apps. AI tools are now helping dentists detect dental decay much earlier, and interdisciplinary collaborations are tackling broader health challenges, such as the impact of climate change on oral health. Researchers are using AI to analyze environmental data and identify vulnerable communities, while AI is also accelerating biomedical research and therapeutic discovery. In dental practices, AI technologies are optimizing imaging analysis and automating administrative tasks. However, experts cautioned about the need for ethical considerations, transparency, and addressing biases in AI systems. The symposium emphasized the importance of integrating human factors like empathy and creativity as AI continues to evolve in the healthcare sector.

🔹 Bullet Summary:
- AI is transforming oral healthcare by improving disease detection and increasing access to care, as discussed at Harvard's symposium.
- AI tools help dentists detect decay earlier and address population-level health challenges, requiring cross-disciplinary collaboration.
- AI in dentistry enhances patient engagement and office efficiency, but ethical concerns about fairness and transparency remain.

🔹 Fallback Summary:
The inaugural Global Symposium on AI & Dentistry at Harvard highlighted the transformative potential of AI in oral healthcare. Experts discussed how AI could revolutionize dental practices by enhancing disease detection, improving patient outcomes, and increasing access to care. AI tools are already aiding dentists in identifying dental issues earlier and automating administrative tasks. Researchers are exploring AI's role in addressing broader health challenges, such as the impact of climate change on oral health. AI is also being used to optimize biomedical research and therapeutic discovery.

However, experts caution that while AI offers exciting possibilities, it lacks human qualities like empathy and creativity, which are crucial for ethical problem-solving. There is a need for transparency and trust in AI systems, especially in healthcare, where decisions can have significant consequences. Guidelines and policies must evolve to address fairness and bias in AI applications. Overall, the symposium emphasized the importance of cross-disciplinary collaboration and careful integration of AI into dental care to maximize its benefits while addressing potential challenges.

📝 Article 9:

🔹 Role Summary:
A recent study from Mass General Brigham (MGB) using a new AI tool has found that 22.8% of the population may suffer from long COVID, significantly higher than previous estimates of 7%. This AI tool analyzes electronic health records to identify long COVID cases by distinguishing symptoms linked to COVID-19 from other conditions. The study, involving data from nearly 300,000 patients across 14 hospitals and 20 community health centers, was published in the journal Med. The tool's "precision phenotyping" method identifies enduring symptoms like fatigue and brain fog, ensuring that only cases not explained by other medical conditions are flagged as long COVID. The AI tool aims to improve diagnosis and care for long COVID by providing a more accurate and unbiased assessment compared to traditional ICD-10 diagnostic codes, which often favor those with better healthcare access. The researchers acknowledge some limitations, such as incomplete health record data and challenges in identifying initial COVID-19 infections due to declining testing. The study was limited to Massachusetts, but future plans include testing the algorithm in diverse patient cohorts and making it publicly available for global use. This work, supported by various national and international research grants, could enhance clinical care and facilitate research into the genetic and biochemical underpinnings

🔹 Bullet Summary:
- A new AI tool by Mass General Brigham identifies long COVID in 22.8% of the population, higher than previous estimates.
- The tool uses "precision phenotyping" to differentiate long COVID symptoms from other conditions in electronic health records.
- The algorithm reduces diagnostic bias, better reflecting diverse populations, but faces limitations in data completeness and testing decline.

🔹 Fallback Summary:
A new AI tool developed by Mass General Brigham has revealed that 22.8% of the population may suffer from long COVID, a significantly higher percentage than previous estimates. This tool analyzes electronic health records to help identify long COVID cases, which can include symptoms like fatigue, chronic cough, and brain fog. The algorithm, which was developed using data from nearly 300,000 patients, aims to improve diagnosis by distinguishing long COVID symptoms from other conditions.

The study highlights that the AI tool can provide more accurate and less biased diagnoses than traditional method

```
s, potentially benefiting marginalized communities. However, there are limitations, such as incomplete health reco
rd data and challenges in identifying initial COVID-19 infections due to reduced testing. The research was conduct
ed in Massachusetts, and future studies may expand its application.

The researchers plan to make the algorithm publicly available for global use, which could enhance clinical care an
d facilitate research into the genetic and biochemical aspects of long COVID. The study received support from vari
ous health institutes and organizations.

📝 Article 10:

🔹 Role Summary:
The article discusses the potential and challenges of using artificial intelligence (AI) in medicine and public he
alth, as highlighted by Andrew Beam from the Harvard T.H. Chan School of Public Health. Beam, along with Arjun (Ra
j) Manrai from Harvard Medical School, is co-deputy editor of the new journal NEJM AI, which will publish its firs
t articles in the fall, and co-hosts the podcast NEJM AI Grand Rounds. The journal and podcast aim to explore AI's
role in medicine, a field where the stakes are particularly high due to the direct impact on patient lives. The fi
rst podcast episodes cover AI's application in genomics, cardiology, radiology, and ophthalmology, with experts li
ke Euan Ashley and Lily Peng discussing AI's capabilities, such as reading echocardiograms and screening for diabe
tic retinopathy. Despite AI's promise, Beam emphasizes the need to address structural biases in AI systems, which
could adversely affect marginalized populations in healthcare.

🔹 Bullet Summary:
- AI has the potential to significantly improve medicine and public health, but caution is necessary.
- NEJM AI journal and podcast explore AI's role in medicine, co-led by Andrew Beam and Arjun Manrai.
- AI's promise in healthcare includes reading echocardiograms and screening for diabetic retinopathy, but biases r
emain a concern.

🔹 Fallback Summary:
Artificial intelligence (AI) has the potential to significantly improve medicine and public health, but it must be
used carefully, according to Andrew Beam from the Harvard T.H. Chan School of Public Health. Beam is involved in t
wo initiatives by NEJM Group to explore AI's role in medicine: the launch of the NEJM AI journal and the NEJM AI G
rand Rounds podcast. These platforms aim to critically evaluate when and how AI can be safely integrated into heal
thcare, as the stakes are particularly high in this field. The podcast's initial episodes cover AI applications in
cardiology, radiology, and ophthalmology, illustrating AI's ability to perform tasks like reading echocardiograms
and screening for diabetic retinopathy effectively. However, Beam also warns about the challenges, such as structu
ral biases in AI systems, which could negatively impact marginalized groups if not addressed.
```

In [16]:

```python
load_dotenv()
client = OpenAI(api_key=os.getenv("OPENAI_API_KEY"))

# 1. Build evaluation prompt
def build_evaluation_prompt(article, summary):
    return f"""
You are an expert reviewer assessing the quality of a summary of a scientific article.

Article:
\"\"\"{article}\"\"\"

Summary:
\"\"\"{summary}\"\"\"

Please rate the summary on a scale from 1 to 5 (5 = best) on the following criteria:

1. Coverage: How well does the summary cover the main points of the article?
2. Conciseness: Is the summary brief and to the point?
3. Coherence: Is the summary logically structured and easy to follow?
4. Faithfulness: Does the summary accurately represent the facts in the article?

Reply only in this format (no explanation, just numbers):
Coverage: <1-5>
Conciseness: <1-5>
Coherence: <1-5>
Faithfulness: <1-5>
"""

# 2. Parse the plain-text response into 4 numeric scores
def parse_scores(text):
    try:
        lines = text.strip().splitlines()
        scores = {}
        for line in lines:
            key, val = line.split(":")
            scores[key.strip().lower()] = int(val.strip())
        return (
            scores.get("coverage", None),
```

```python
            scores.get("conciseness", None),
            scores.get("coherence", None),
            scores.get("faithfulness", None)
        )
    except Exception as e:
        # If parsing fails, return None
        return None, None, None, None

# 3. Evaluate a single summary
def evaluate_summary_llm(article, summary, model="gpt-4o"):
    if summary.startswith("⚠️") or len(summary.strip()) < 20:
        return None, None, None, None

    prompt = build_evaluation_prompt(article, summary)

    try:
        response = client.chat.completions.create(
            model=model,
            messages=[
                {"role": "system", "content": "You are an expert summary evaluator."},
                {"role": "user", "content": prompt}
            ],
            temperature=0,
            max_tokens=100
        )
        raw_output = response.choices[0].message.content
        return parse_scores(raw_output)
    except Exception as e:
        return None, None, None, None

# 4. Add evaluation scores to the DataFrame for one summary column
def add_scores_to_df(df, summary_col):
    covs, concs, cohes, faiths = [], [], [], []

    for i, row in df.iterrows():
        article = row["content"]
        summary = row[summary_col]
        c, co, ch, f = evaluate_summary_llm(article, summary)
        covs.append(c)
        concs.append(co)
        cohes.append(ch)
        faiths.append(f)

    df[f"{summary_col}_coverage"] = covs
    df[f"{summary_col}_conciseness"] = concs
    df[f"{summary_col}_coherence"] = cohes
    df[f"{summary_col}_faithfulness"] = faiths
    return df

# 5. Apply to each summary column (edit as needed)
summary_columns = ["summary_role", "summary_bullets", "summary_fallback"]

for col in summary_columns:
    print(f"🔍 Evaluating summaries in column: {col}")
    df = add_scores_to_df(df, col)
```

```
🔍 Evaluating summaries in column: summary_role
🔍 Evaluating summaries in column: summary_bullets
🔍 Evaluating summaries in column: summary_fallback
```

In [17]:

```python
# List of summary strategies
summary_cols = ["summary_role", "summary_bullets", "summary_fallback"]

# Initialize dictionary to store averages
averages = {}

for col in summary_cols:
    averages[col] = {
        "coverage": df[f"{col}_coverage"].mean(),
        "conciseness": df[f"{col}_conciseness"].mean(),
        "coherence": df[f"{col}_coherence"].mean(),
        "faithfulness": df[f"{col}_faithfulness"].mean(),
    }

# Convert to DataFrame for a nice table
```

```
avg_df = pd.DataFrame(averages).T.round(2)

print("\n📊 Average evaluation scores per summary type:")
print(avg_df)
```

```
📊 Average evaluation scores per summary type:
                  coverage  conciseness  coherence  faithfulness
summary_role           4.6          4.9        4.9           4.9
summary_bullets        3.2          4.2        4.2           4.1
summary_fallback       4.1          4.7        4.7           4.7
```

In [18]:
```
structured_output = []

for _, row in df.iterrows():
    record = {
        "title": row.get("title", ""),  # optional, if you have titles
        "summaries": {
            "role": {
                "coverage": row.get("summary_role_coverage"),
                "conciseness": row.get("summary_role_conciseness"),
                "coherence": row.get("summary_role_coherence"),
                "faithfulness": row.get("summary_role_faithfulness"),
            },
            "bullets": {
                "coverage": row.get("summary_bullets_coverage"),
                "conciseness": row.get("summary_bullets_conciseness"),
                "coherence": row.get("summary_bullets_coherence"),
                "faithfulness": row.get("summary_bullets_faithfulness"),
            },
            "fallback": {
                "coverage": row.get("summary_fallback_coverage"),
                "conciseness": row.get("summary_fallback_conciseness"),
                "coherence": row.get("summary_fallback_coherence"),
                "faithfulness": row.get("summary_fallback_faithfulness"),
            }
        }
    }
    structured_output.append(record)

# Save to JSON file
with open("evaluated_summaries.json", "w", encoding="utf-8") as f:
    json.dump(structured_output, f, indent=2, ensure_ascii=False)

print("✅ JSON file 'evaluated_summaries.json' created with structured scores.")
```

```
✅ JSON file 'evaluated_summaries.json' created with structured scores.
```

Reflection & Lessons Learned

This project illustrates the power and modularity of modern NLP pipelines when integrated with LLMs and effective prompt engineering. Several key insights emerged:

- Prompt Engineering is essential: Output quality across tasks (classification, QA, summarization) was directly influenced by the specificity, structure, and tone of the prompts. Even minor changes in phrasing led to significantly different model behaviors.

- Model choice matters: GPT-4o consistently outperformed earlier models like GPT-2 in factuality, coherence, and readability, especially in complex summarization and QA tasks. This supports a trend toward using newer, instruction, aligned models for high-stakes domains like healthcare.

- Zero-shot classification is fast but imperfect: While NLI-based classification offered good generalization, low confidence scores suggest that fine-tuned models or domain-specific taxonomies may yield better granularity.

- RAG systems scale well: The combination of FAISS indexing and GPT-based generation proved effective and lightweight. Embedding-based retrieval ensured contextual relevance without the overhead of large-scale document search engines.

Limitations & Future Improvements

- Dataset size: With only 10 articles, the system could not explore advanced indexing or document clustering. Scaling to hundreds of sources would provide richer insights.

- Evaluation automation: While GPT-based evaluation was informative, incorporating human judgment or ROUGE/BERTScore metrics could improve robustness.
- Interactive front-end: A user interface to allow live querying, exploration of themes, and side-by-side comparison of summaries would greatly enhance usability.
- Summary fusion: Combining multiple summarization strategies (e.g., bullet + role) could produce hybrid outputs optimized for different audiences. For example, a hybrid summary could begin with a high-level bullet-point overview for quick skimming, followed by a more in-depth paragraph capturing nuanced insights. This fusion would cater to both general readers looking for clarity and specialists seeking detailed interpretations, ultimately improving accessibility and engagement across user profiles.